

BFS Traversal

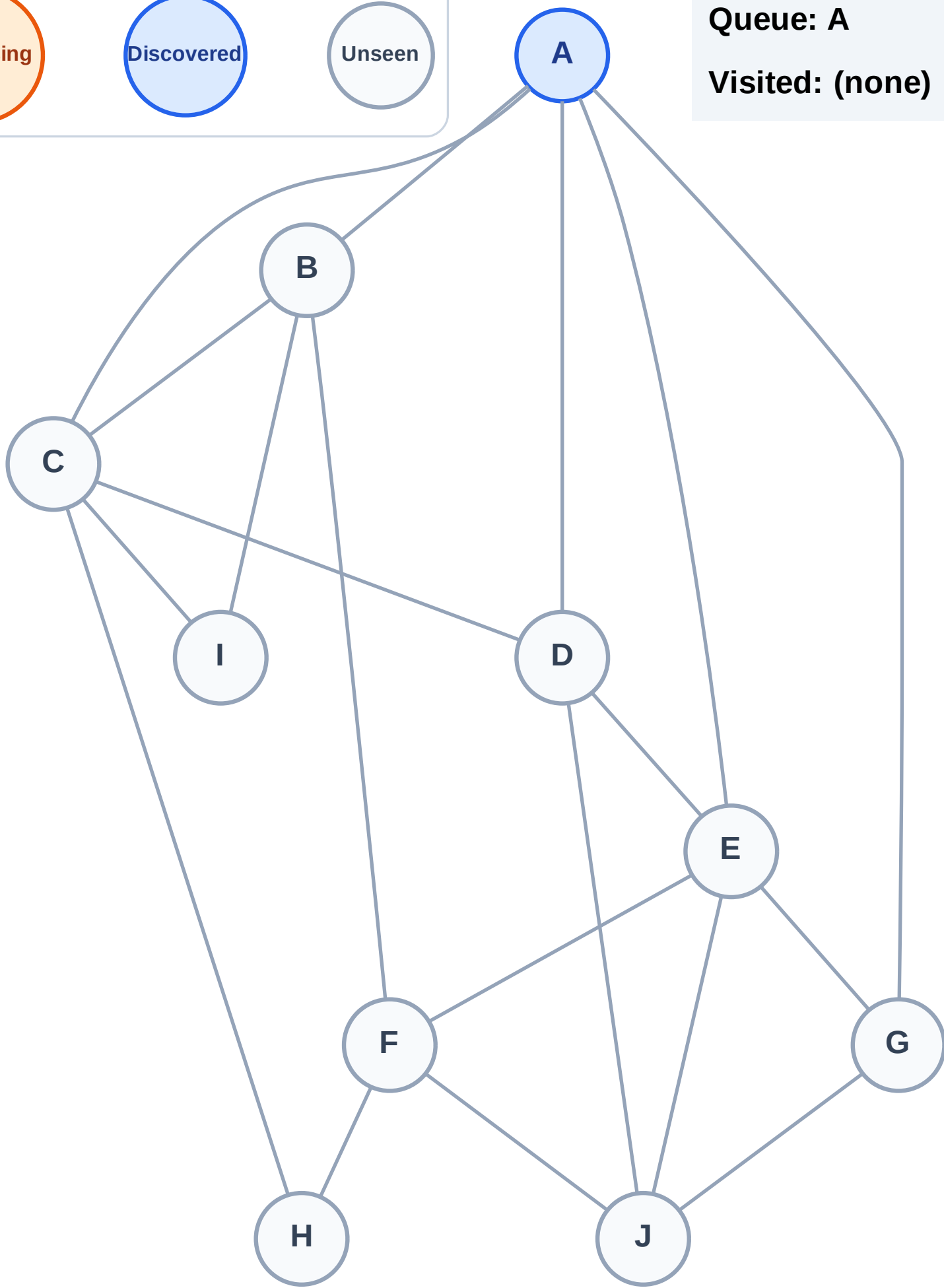
Step 1: Start at A

Legend



Queue: A

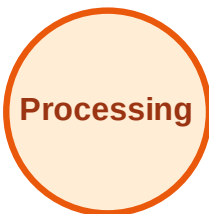
Visited: (none)



BFS Traversal

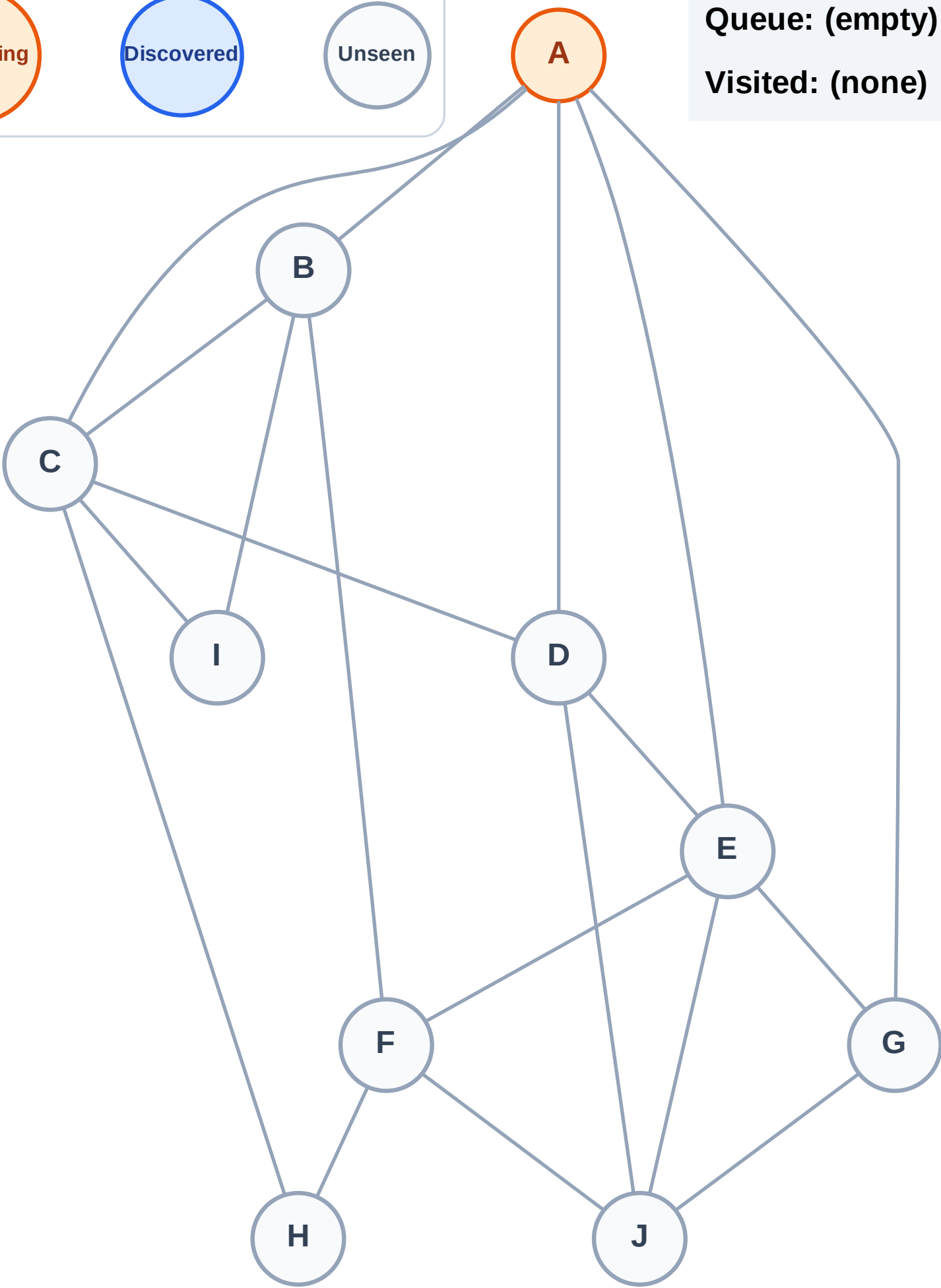
Step 2: Process node A

Legend



Queue: (empty)

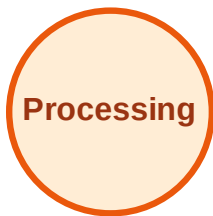
Visited: (none)



BFS Traversal

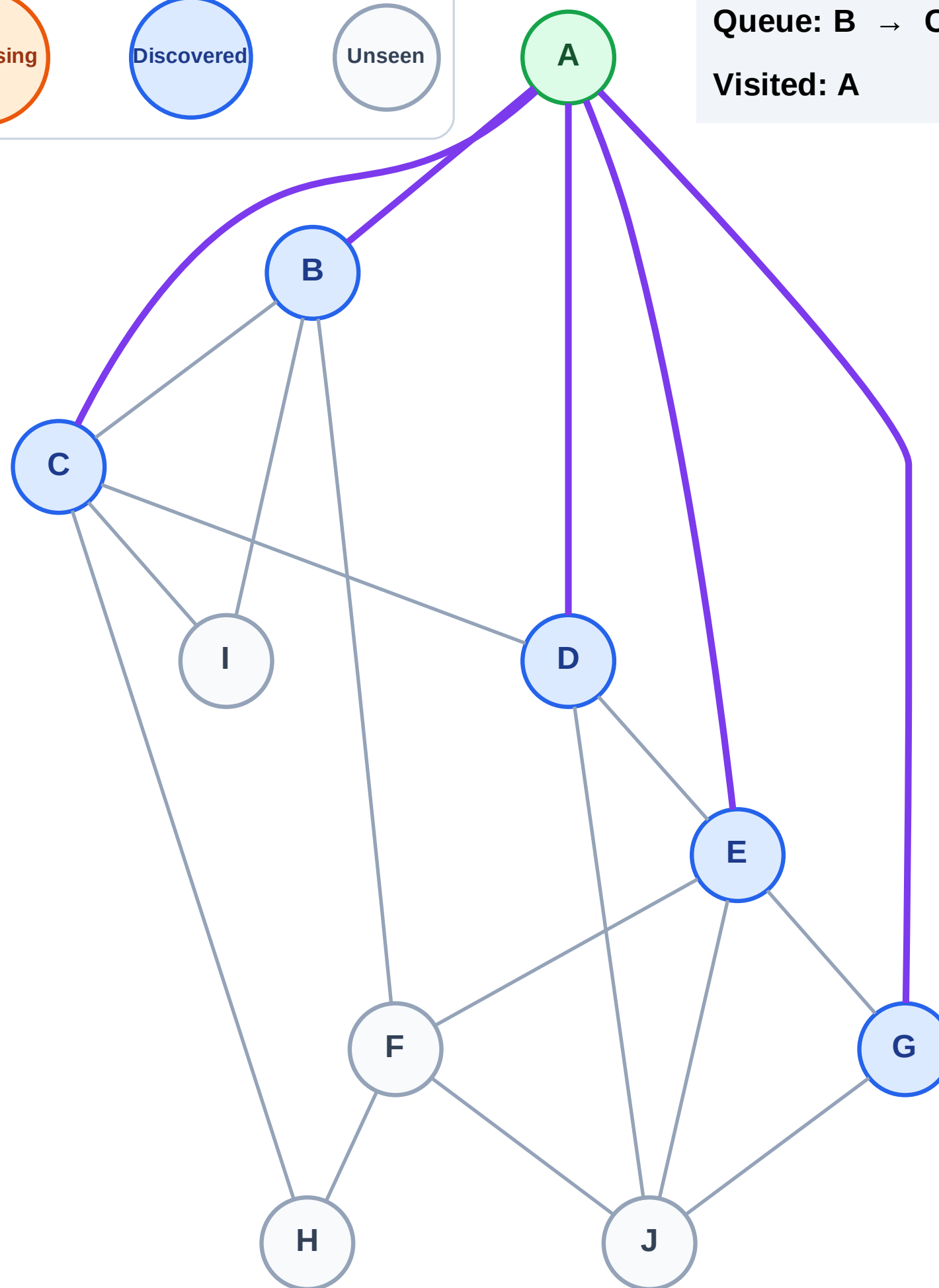
Step 3: Discover B, C, D, E, G from A

Legend



Queue: B → C → D → E → G

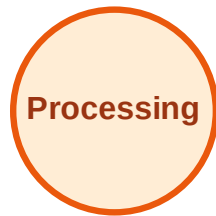
Visited: A



BFS Traversal

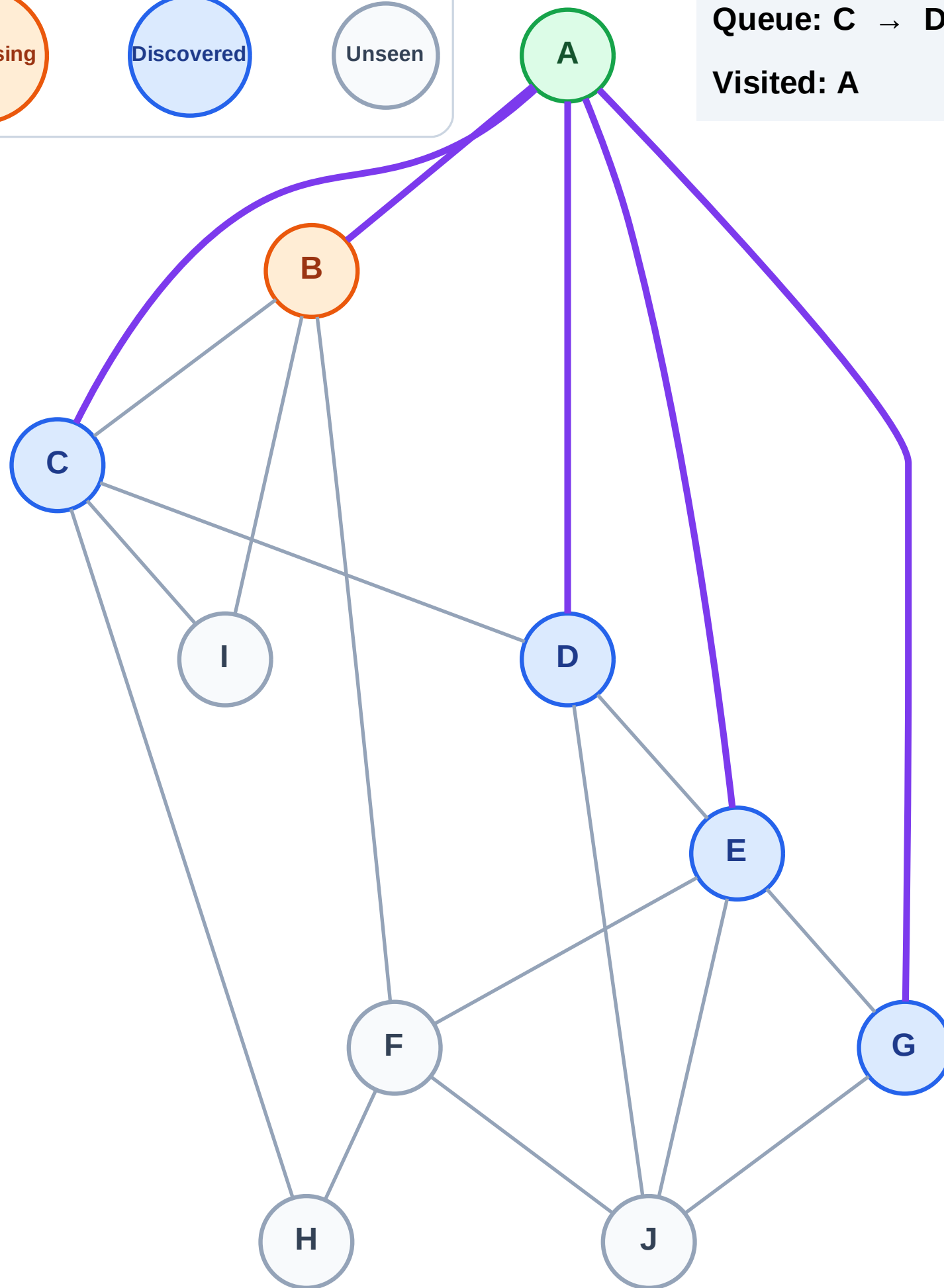
Step 4: Process node B

Legend



Queue: C → D → E → G

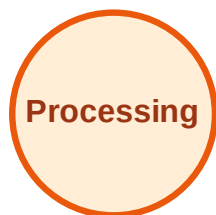
Visited: A



BFS Traversal

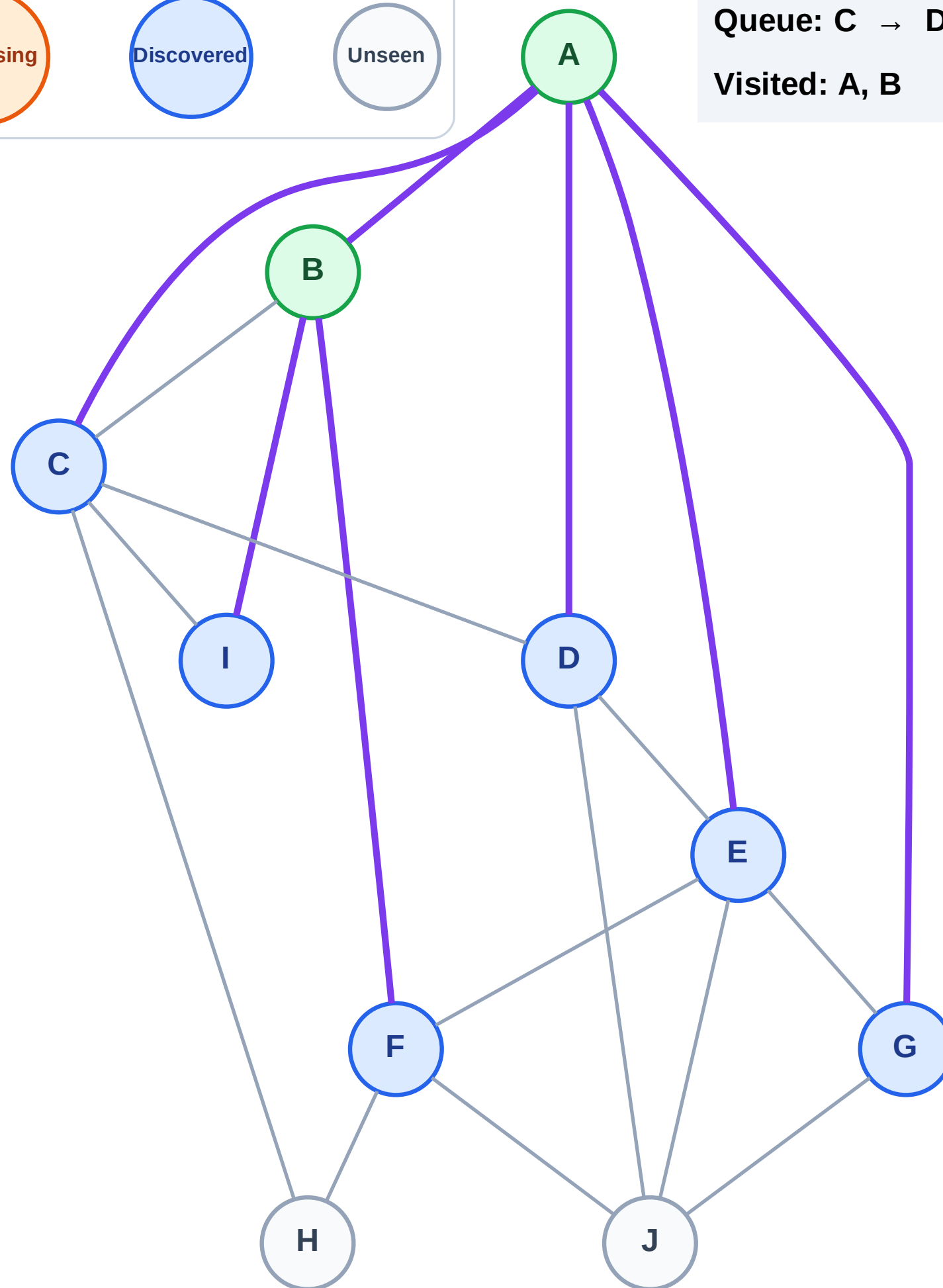
Step 5: Discover F, I from B

Legend



Queue: C → D → E → G → F → I

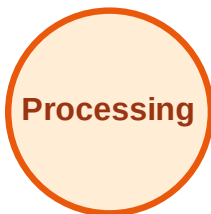
Visited: A, B



BFS Traversal

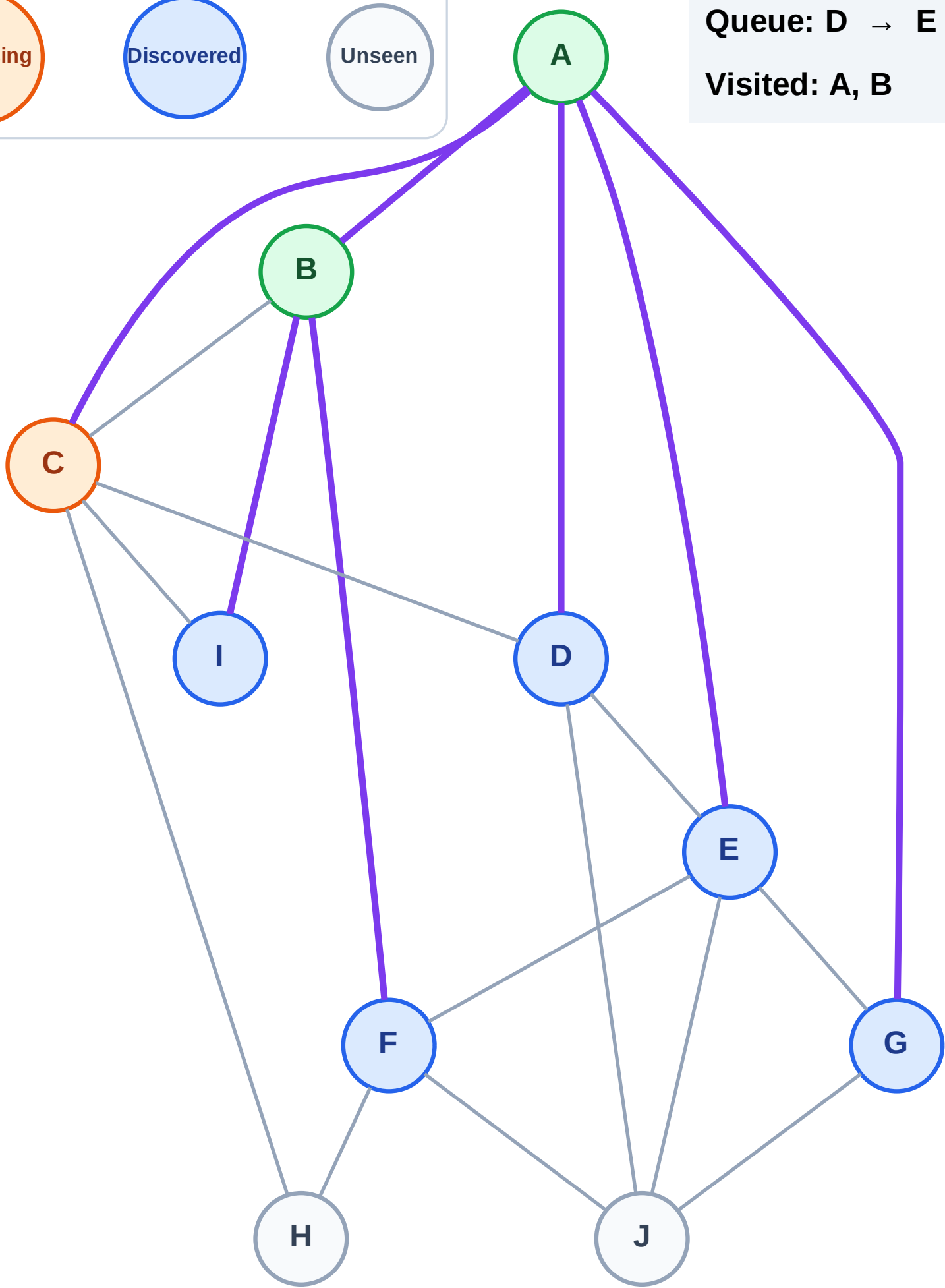
Step 6: Process node C

Legend



Queue: D → E → G → F → I

Visited: A, B



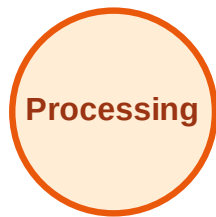
BFS Traversal

Step 7: Discover H from C

Legend



Complete



Processing



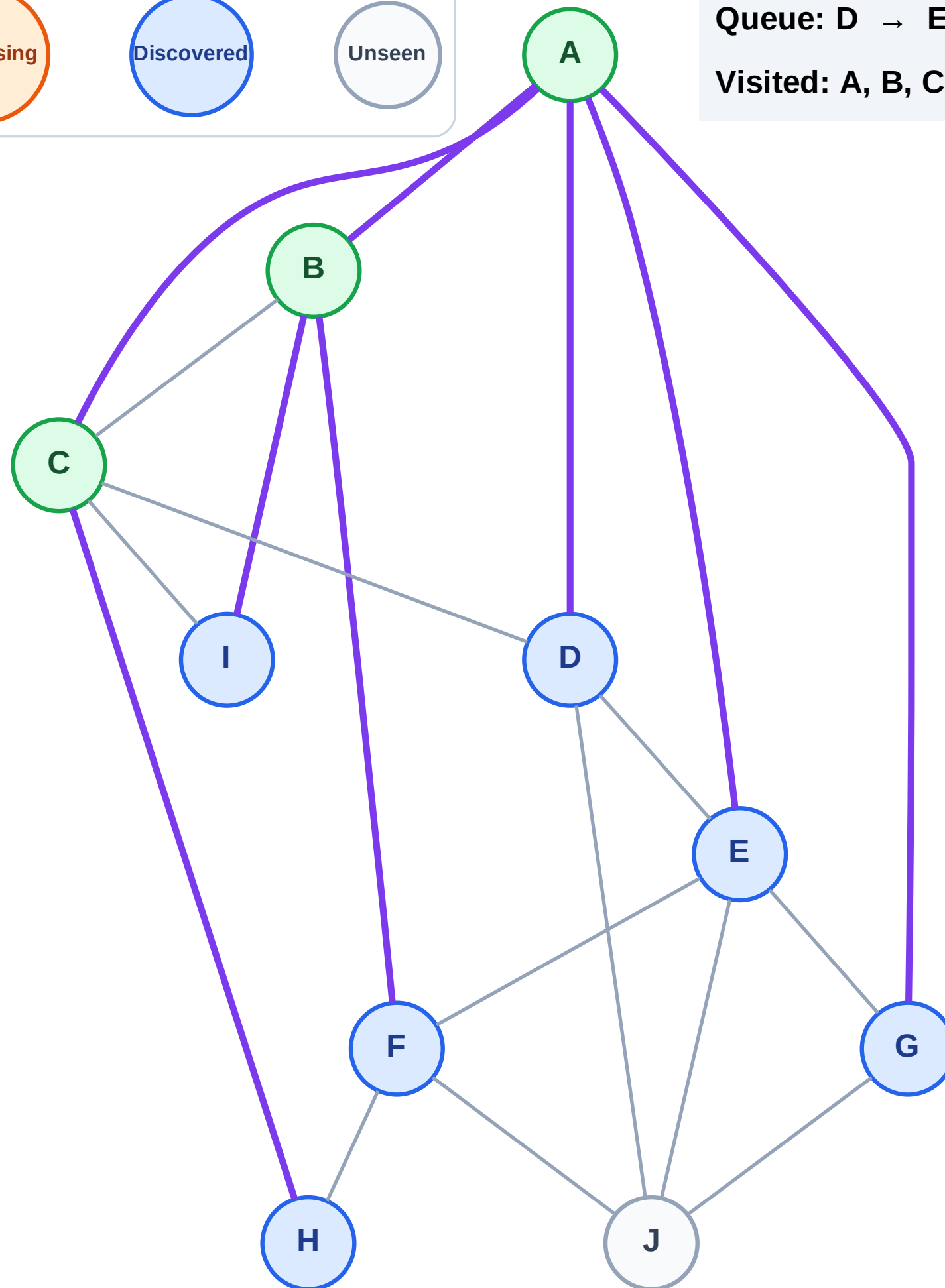
Discovered



Unseen

Queue: D → E → G → F → I → H

Visited: A, B, C



BFS Traversal

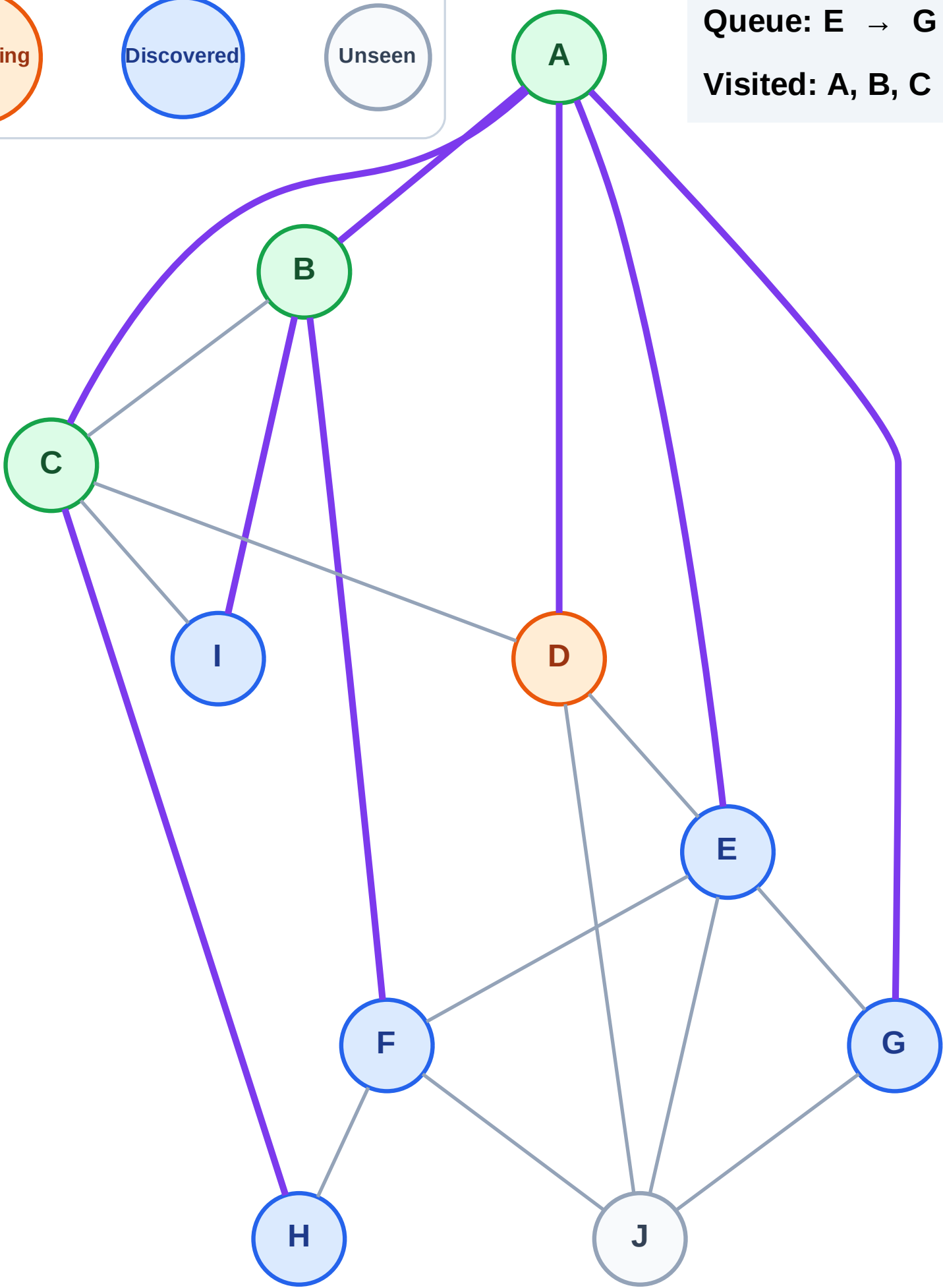
Step 8: Process node D

Legend



Queue: E → G → F → I → H

Visited: A, B, C



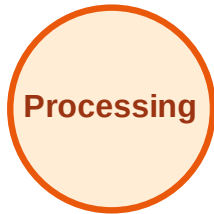
BFS Traversal

Step 9: Discover J from D

Legend



Complete



Processing



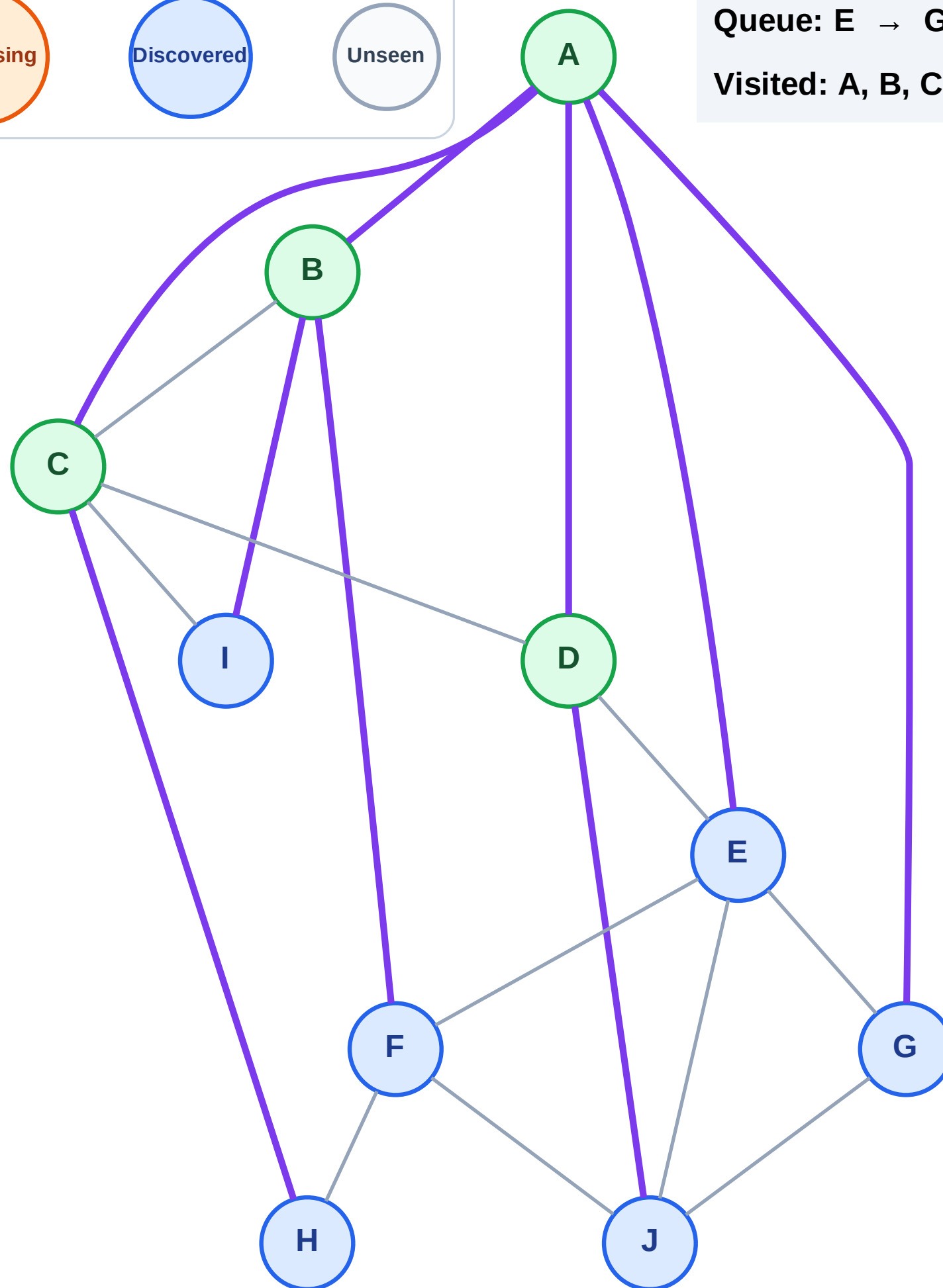
Discovered



Unseen

Queue: E → G → F → I → H → J

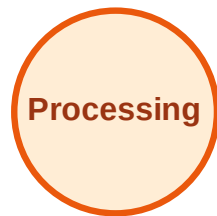
Visited: A, B, C, D



BFS Traversal

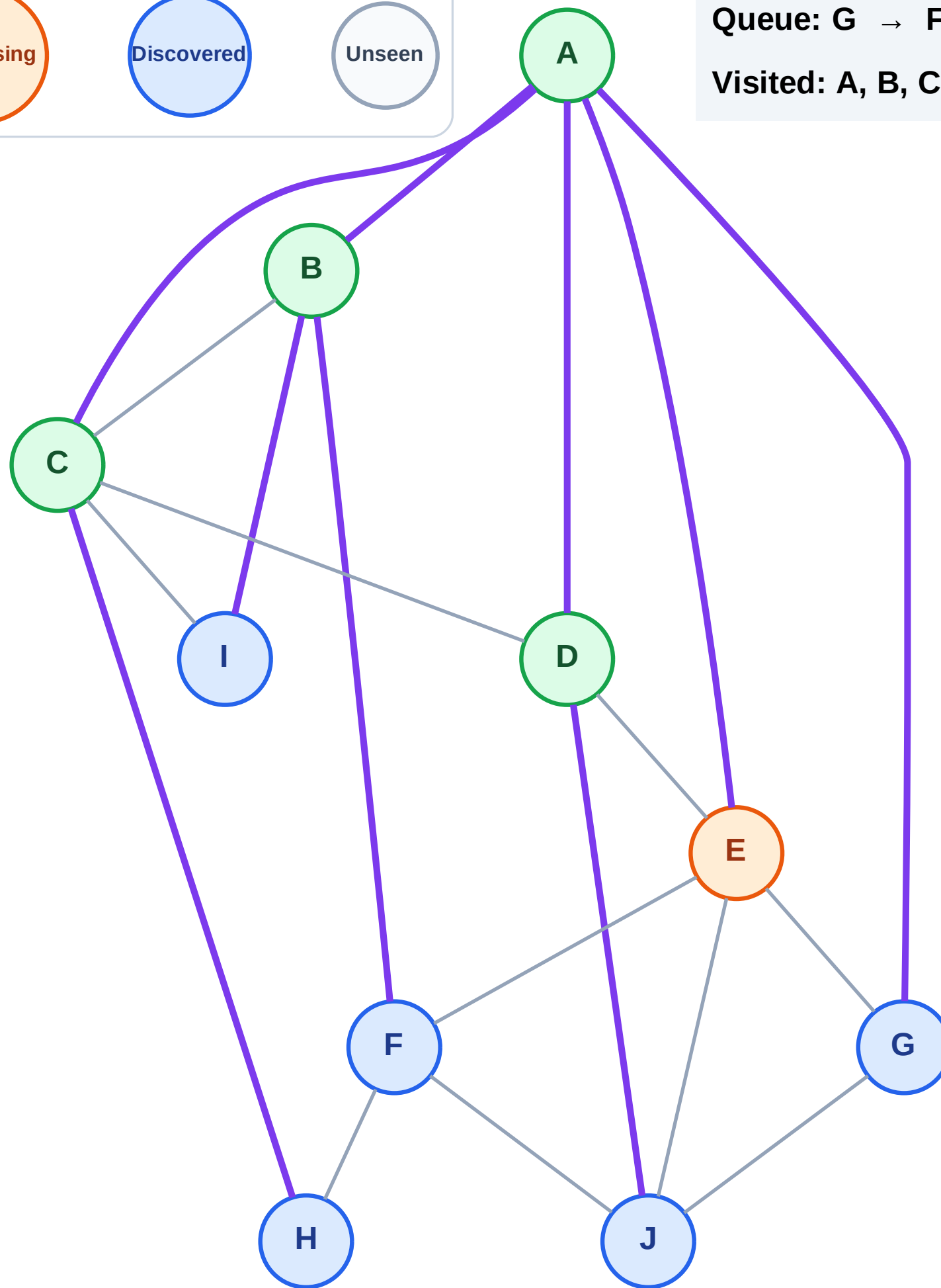
Step 10: Process node E

Legend



Queue: G → F → I → H → J

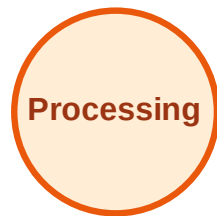
Visited: A, B, C, D



BFS Traversal

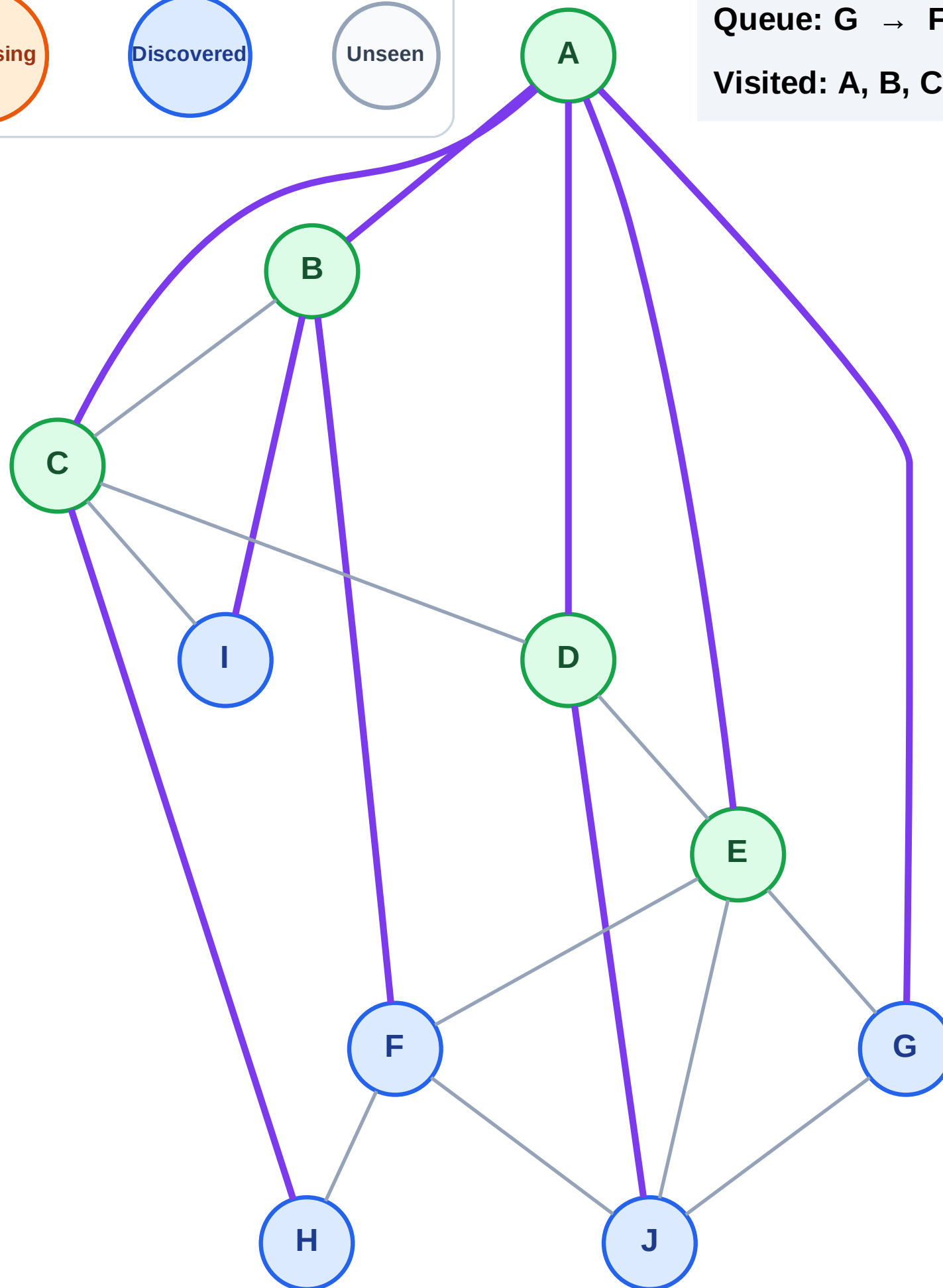
Step 11: No new neighbors from E

Legend



Queue: G → F → I → H → J

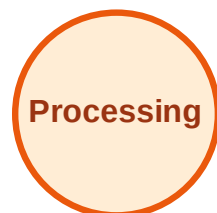
Visited: A, B, C, D, E



BFS Traversal

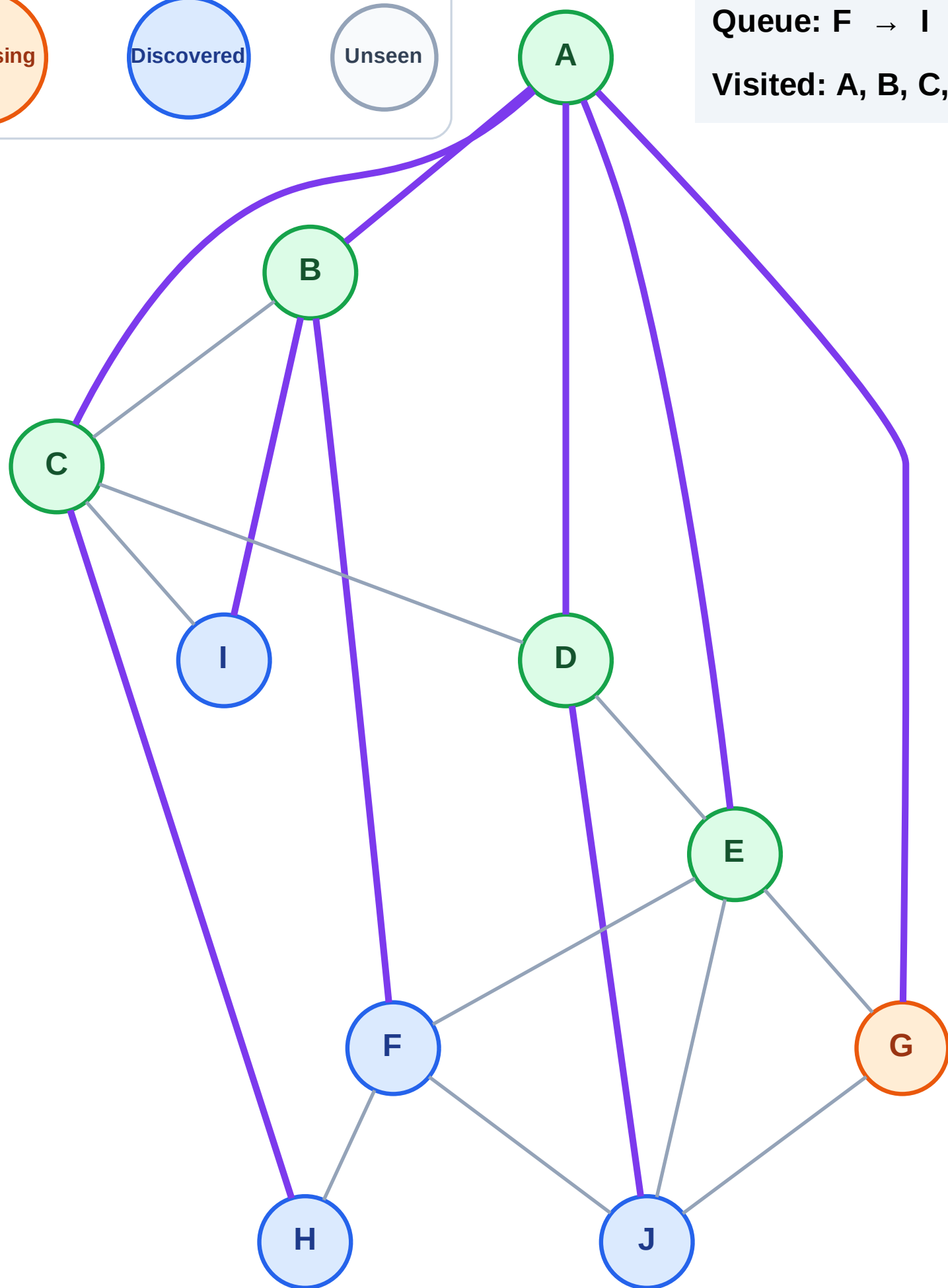
Step 12: Process node G

Legend



Queue: F → I → H → J

Visited: A, B, C, D, E



BFS Traversal

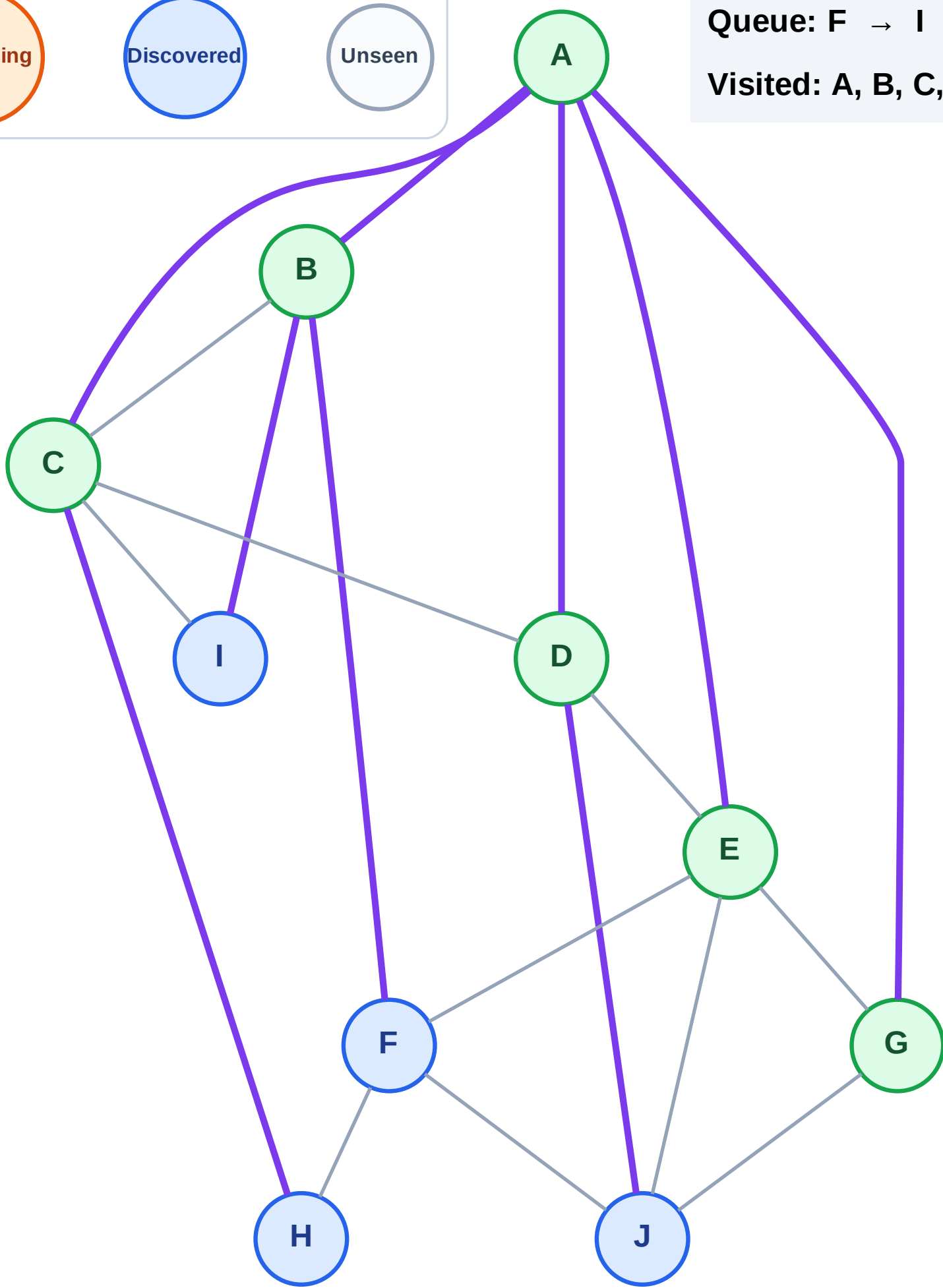
Step 13: No new neighbors from G

Legend



Queue: F → I → H → J

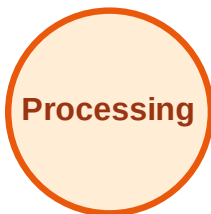
Visited: A, B, C, D, E, G



BFS Traversal

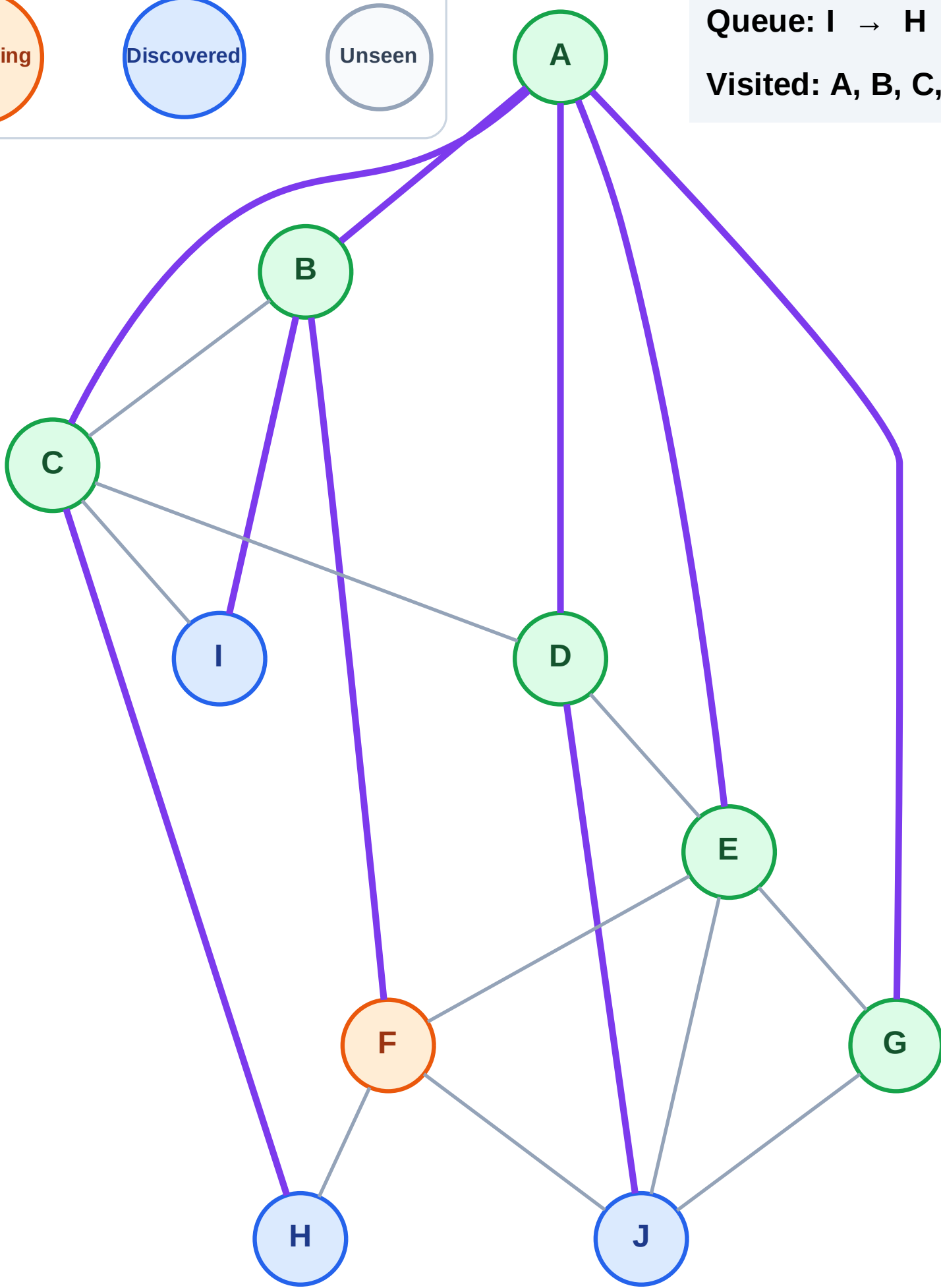
Step 14: Process node F

Legend



Queue: I → H → J

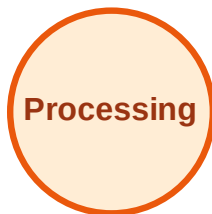
Visited: A, B, C, D, E, G



BFS Traversal

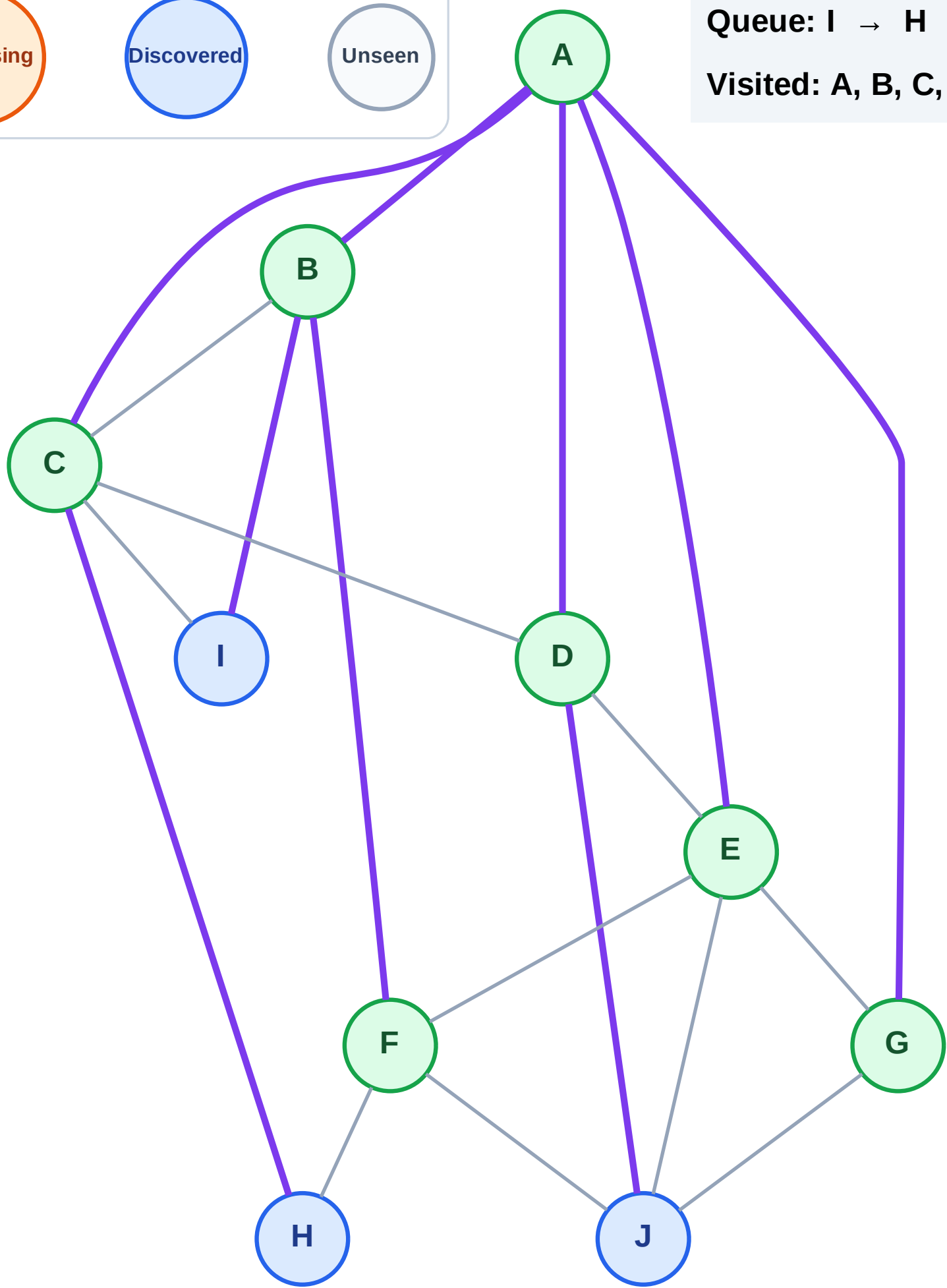
Step 15: No new neighbors from F

Legend



Queue: I → H → J

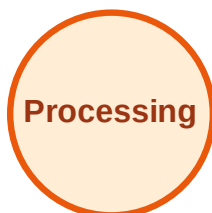
Visited: A, B, C, D, E, F, G



BFS Traversal

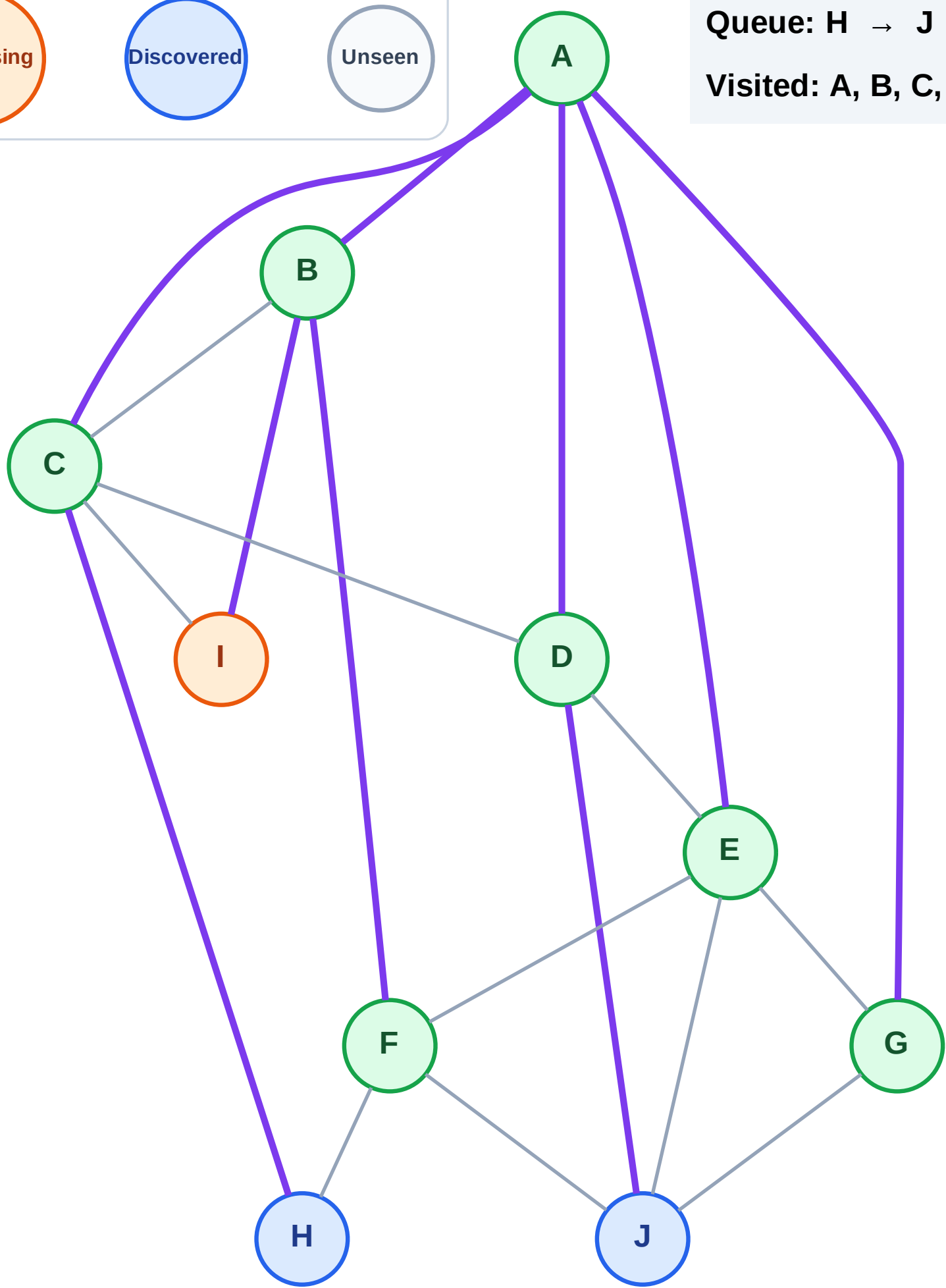
Step 16: Process node I

Legend



Queue: H → J

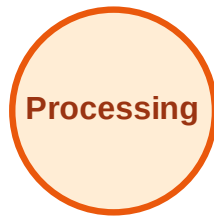
Visited: A, B, C, D, E, F, G



BFS Traversal

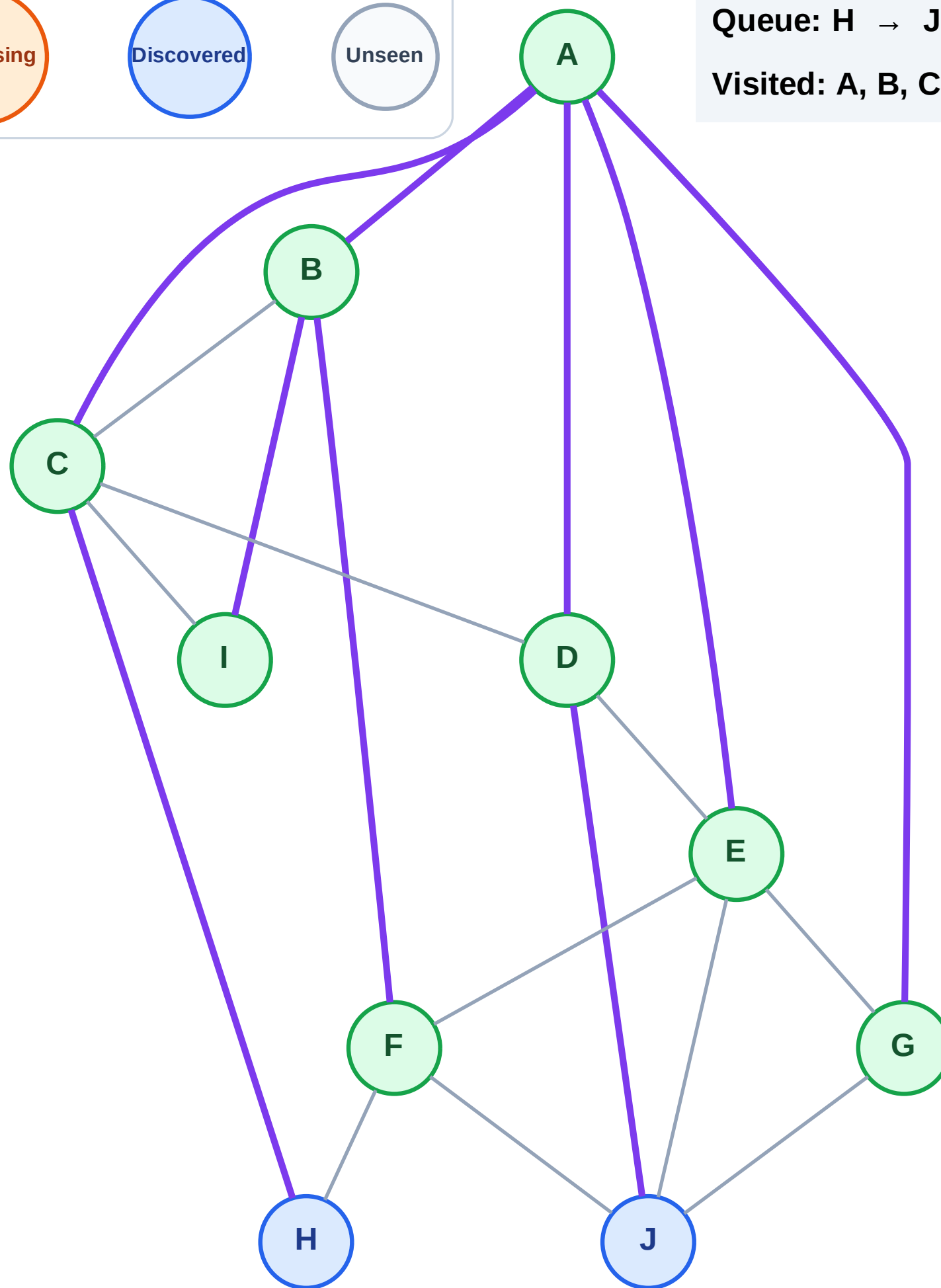
Step 17: No new neighbors from I

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

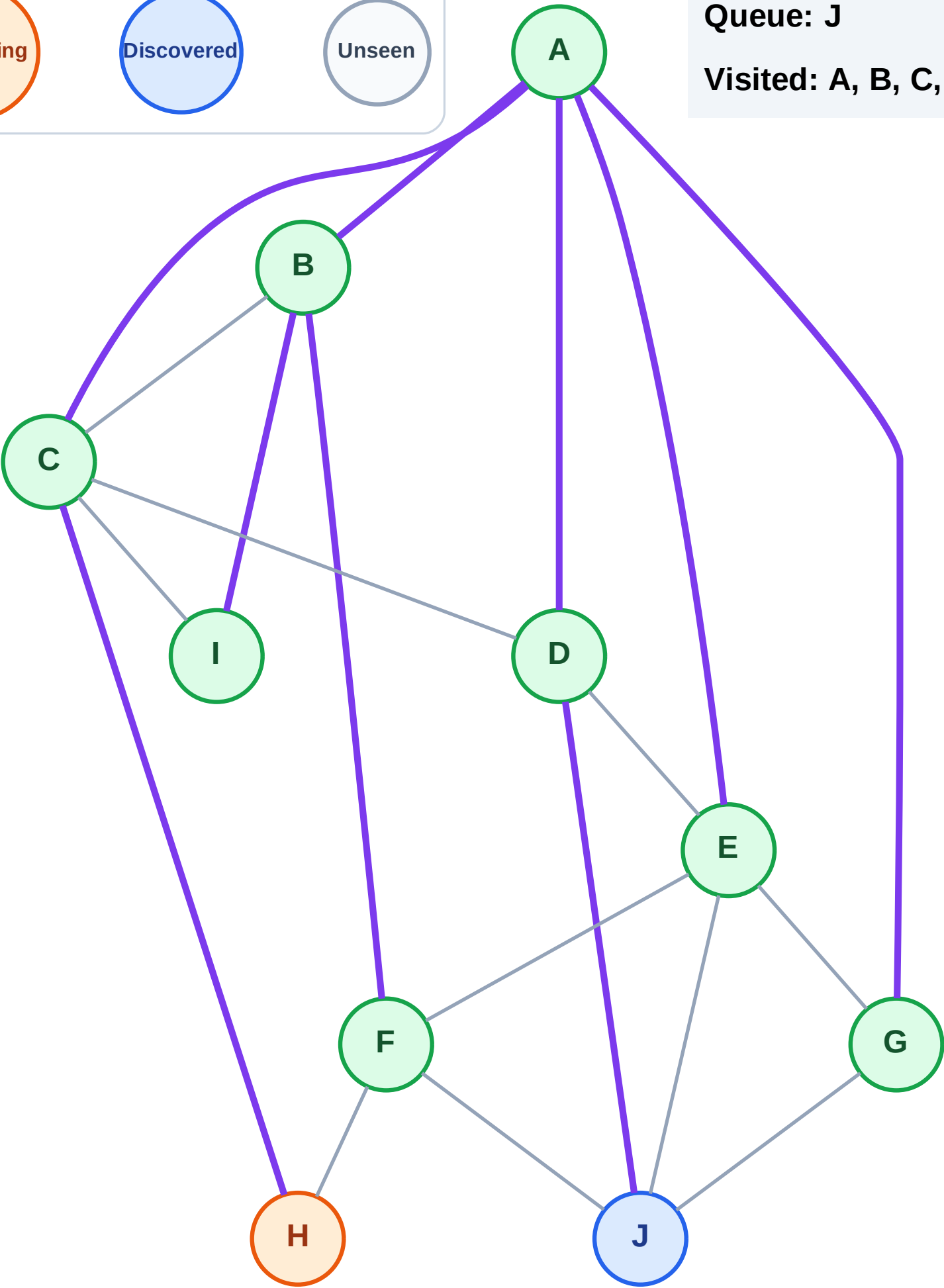
Processing

Discovered

Unseen

Queue: J

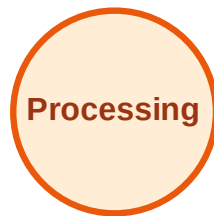
Visited: A, B, C, D, E, F, G, I



BFS Traversal

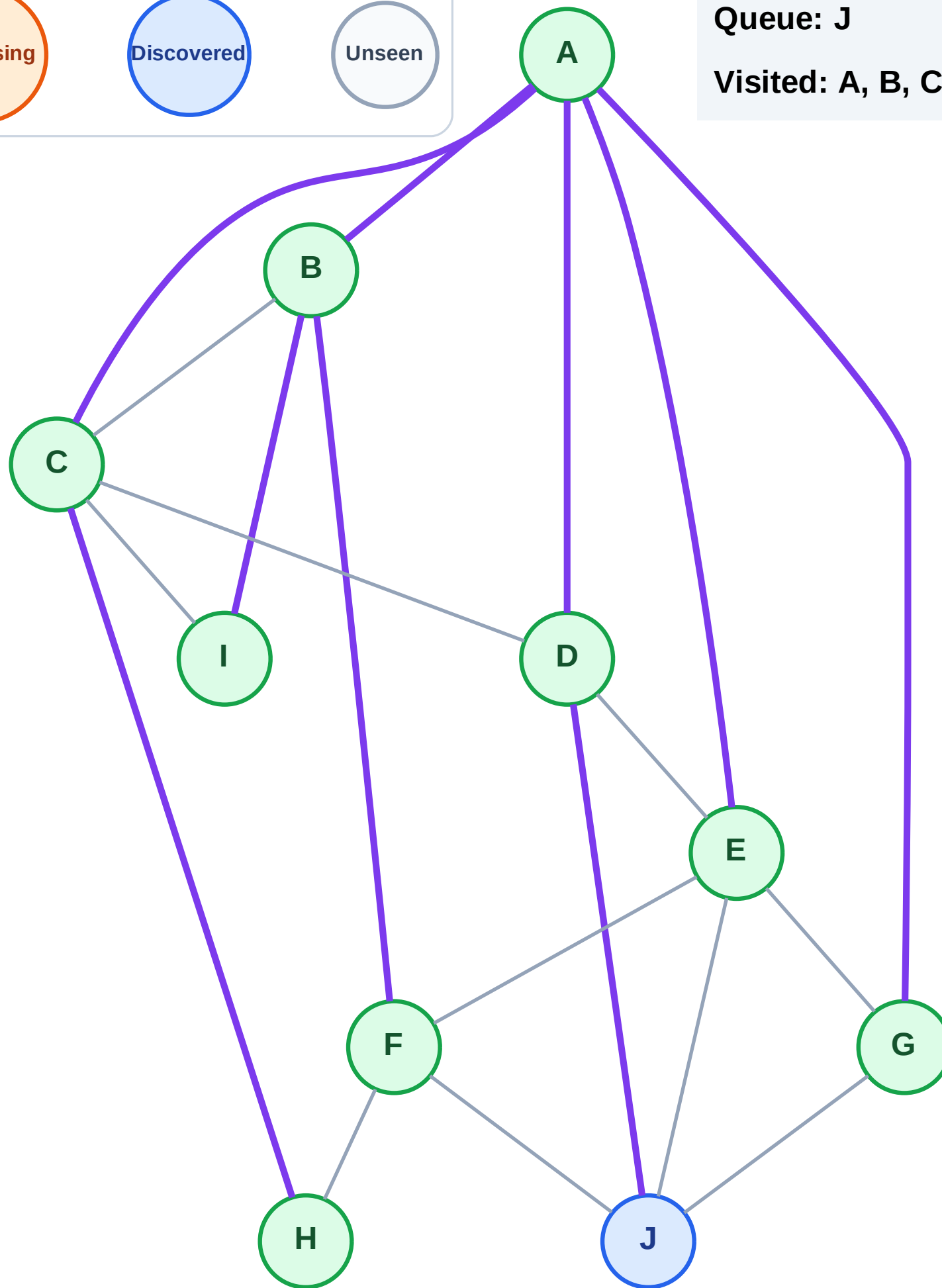
Step 19: No new neighbors from H

Legend



Queue: J

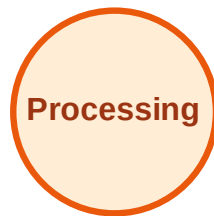
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

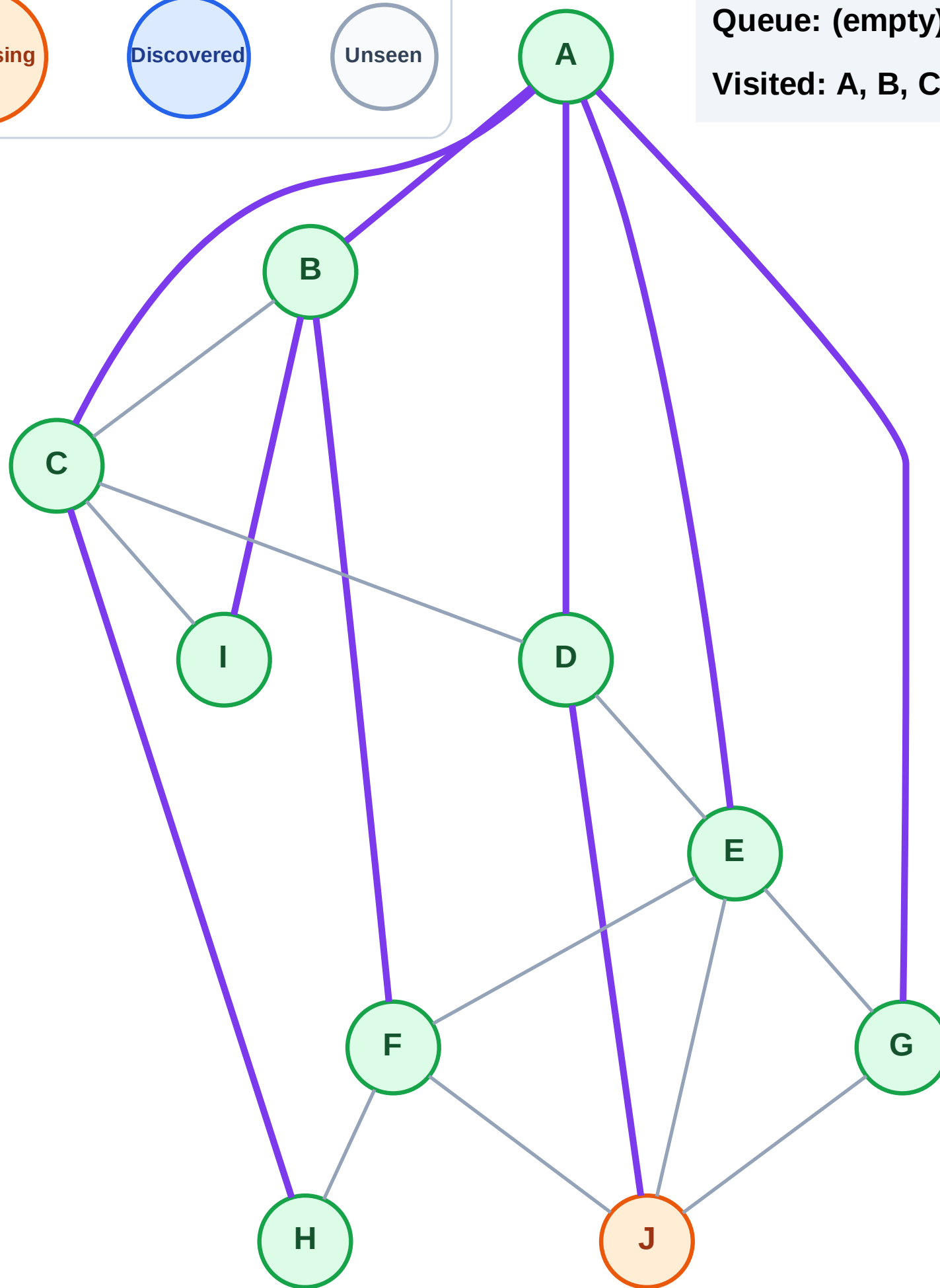
Step 20: Process node J

Legend



Queue: (empty)

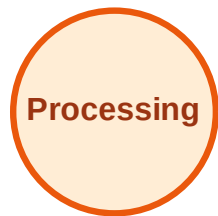
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

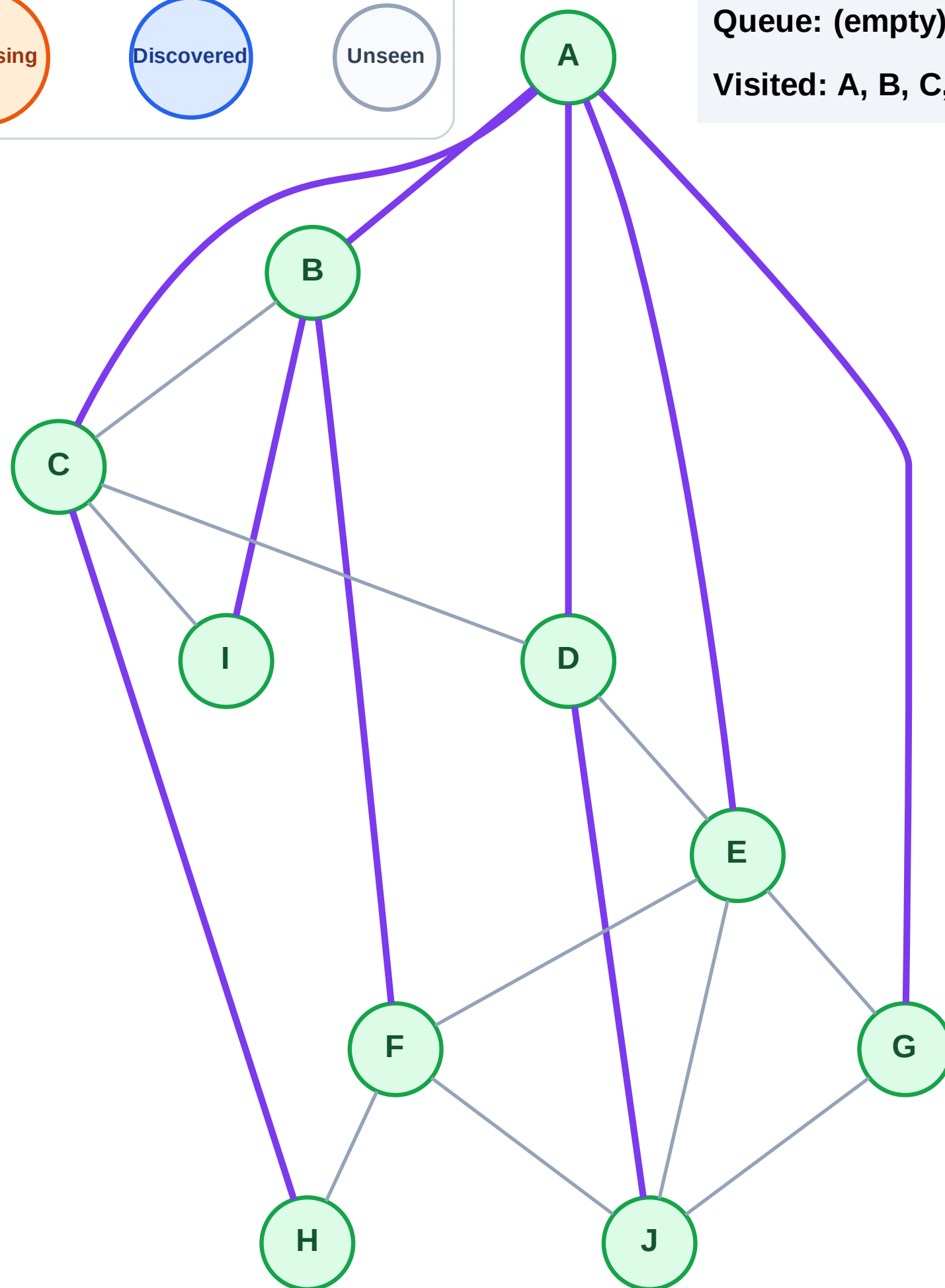
Step 21: No new neighbors from J

Legend



Queue: (empty)

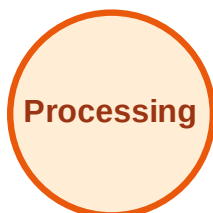
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

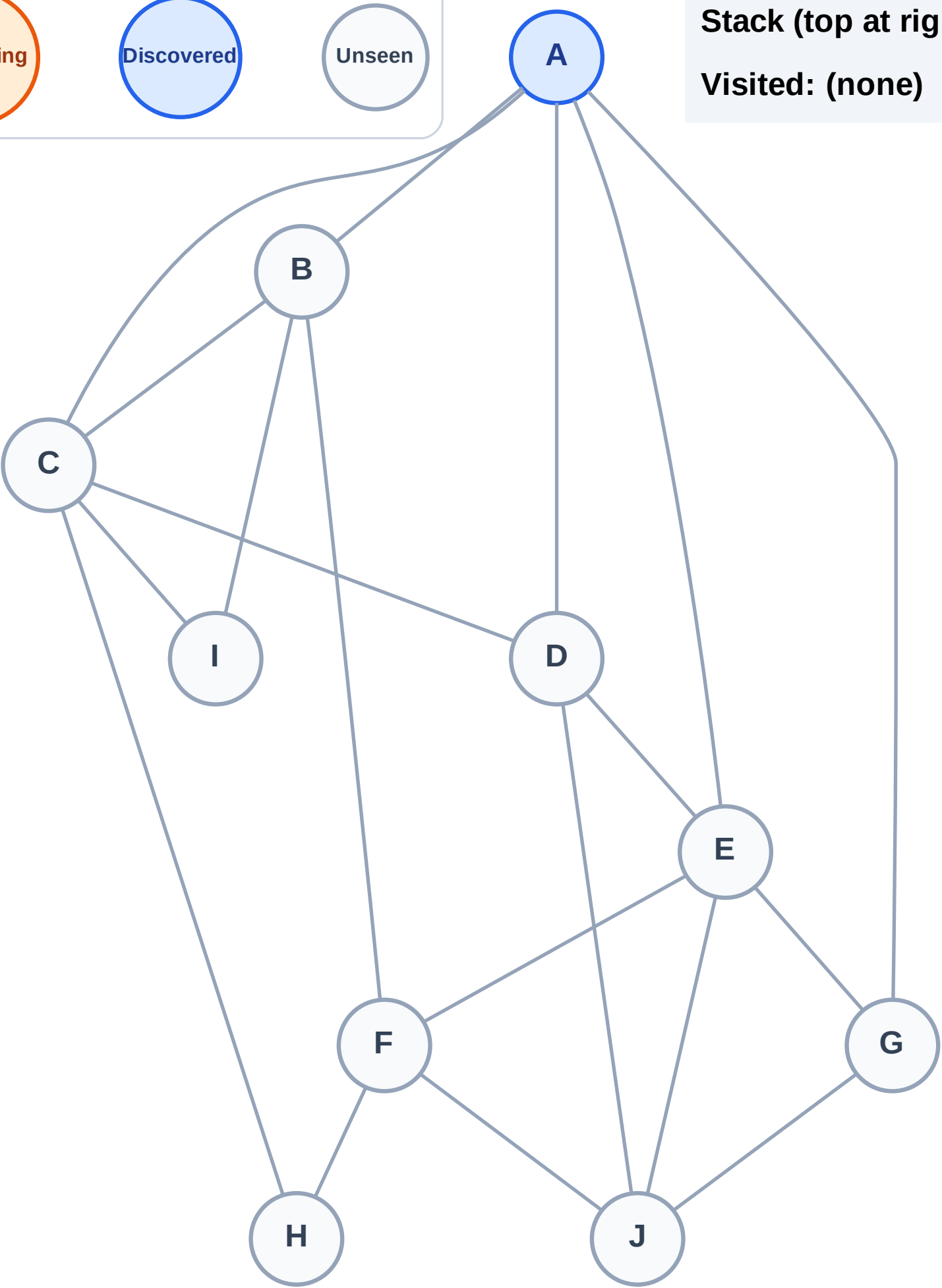
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

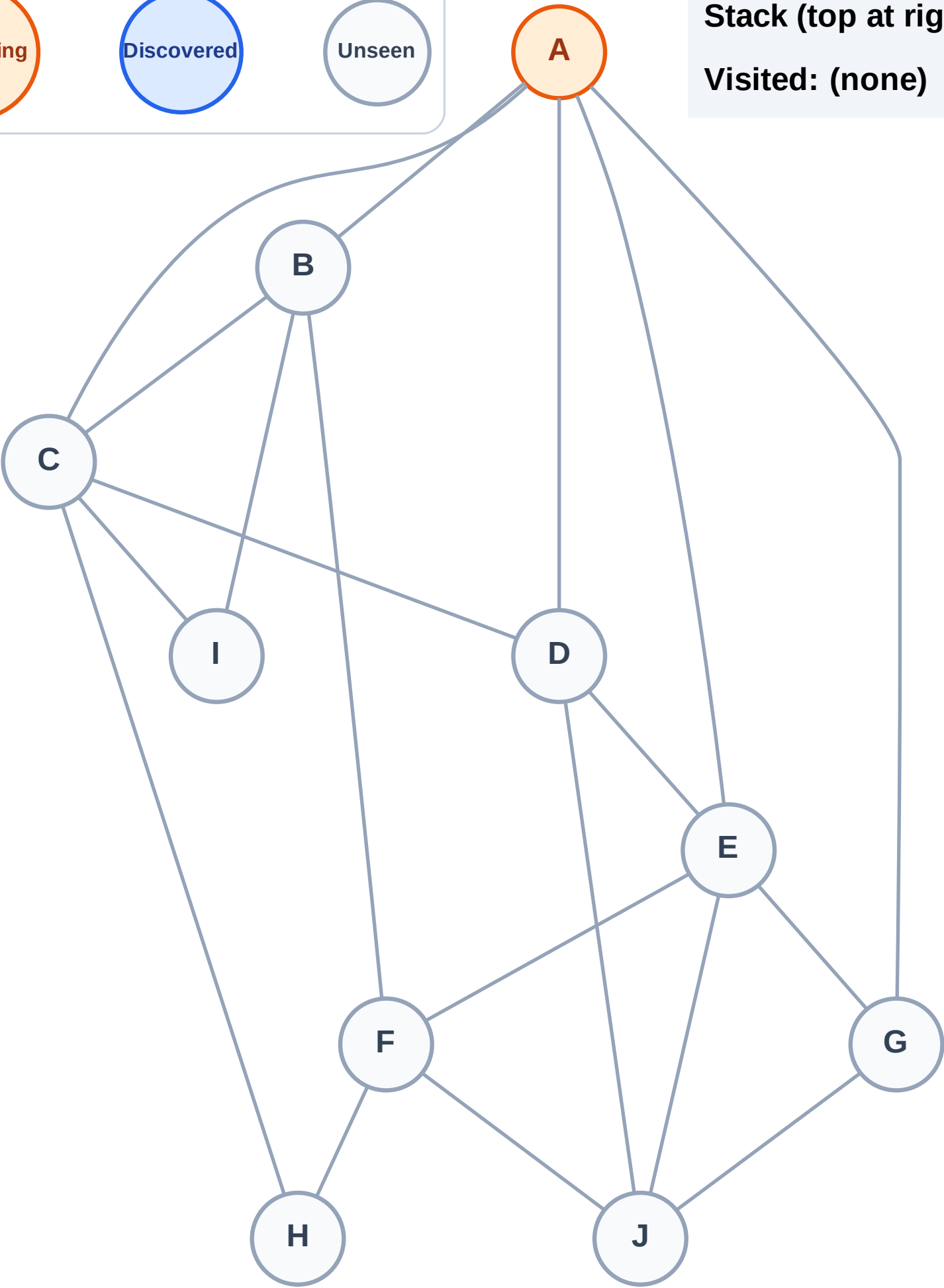
Complete

Processing

Discovered

Unseen

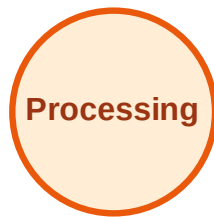
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

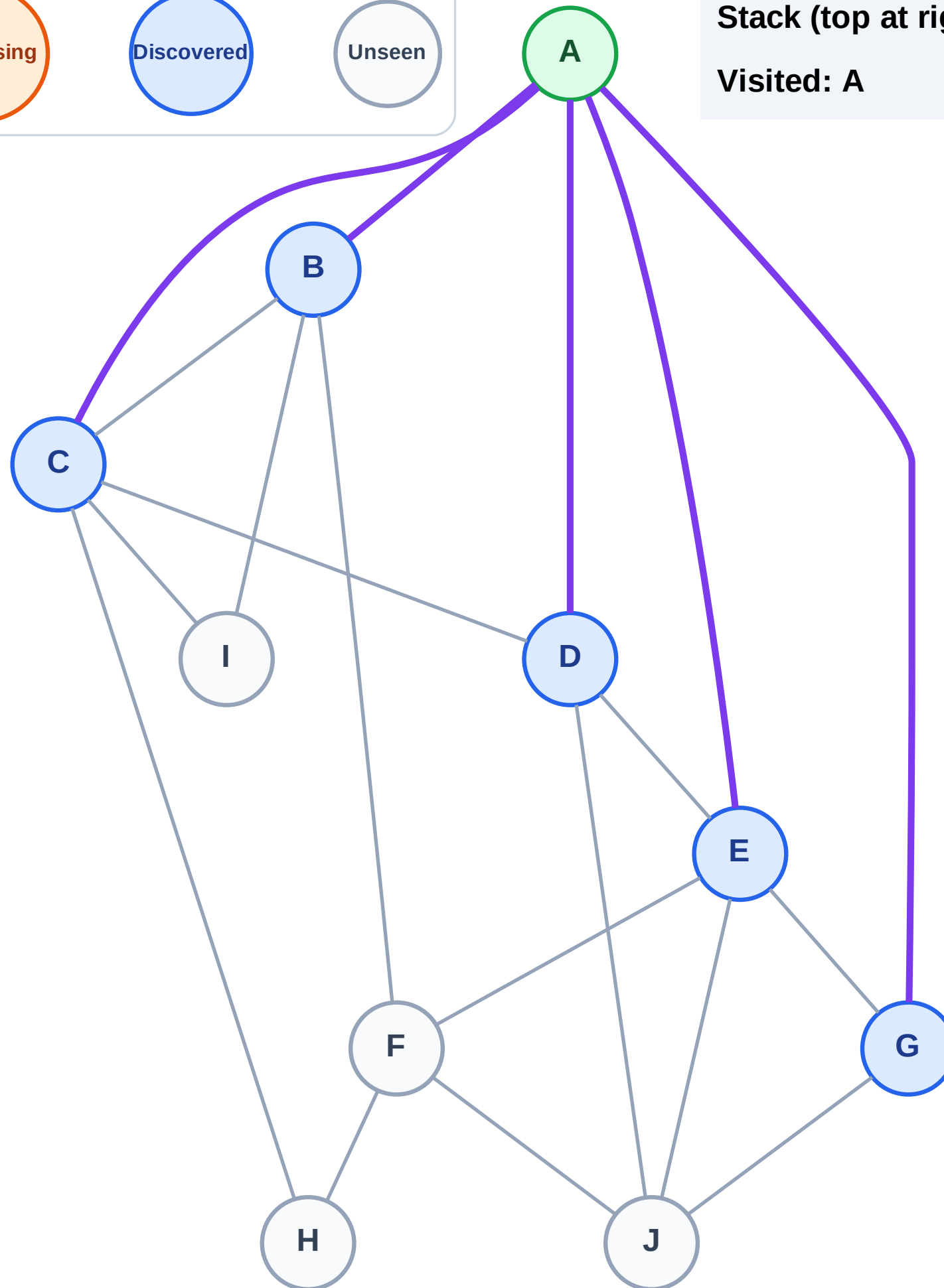
Step 3: Discover G, E, D, C, B from A

Legend



Stack (top at right): G | E | D | C | B

Visited: A



DFS Traversal

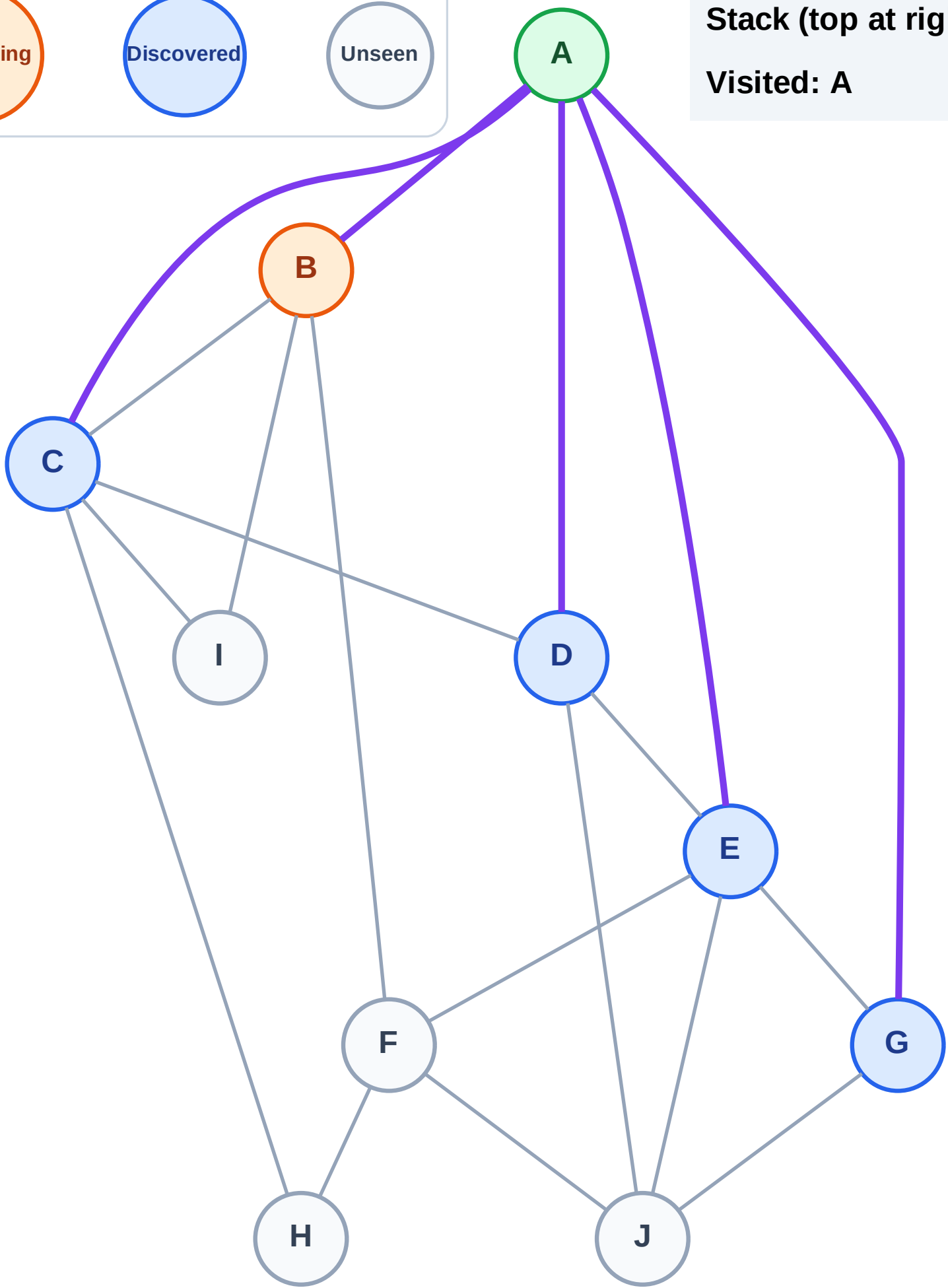
Step 4: Process node B

Legend



Stack (top at right): G | E | D | C

Visited: A

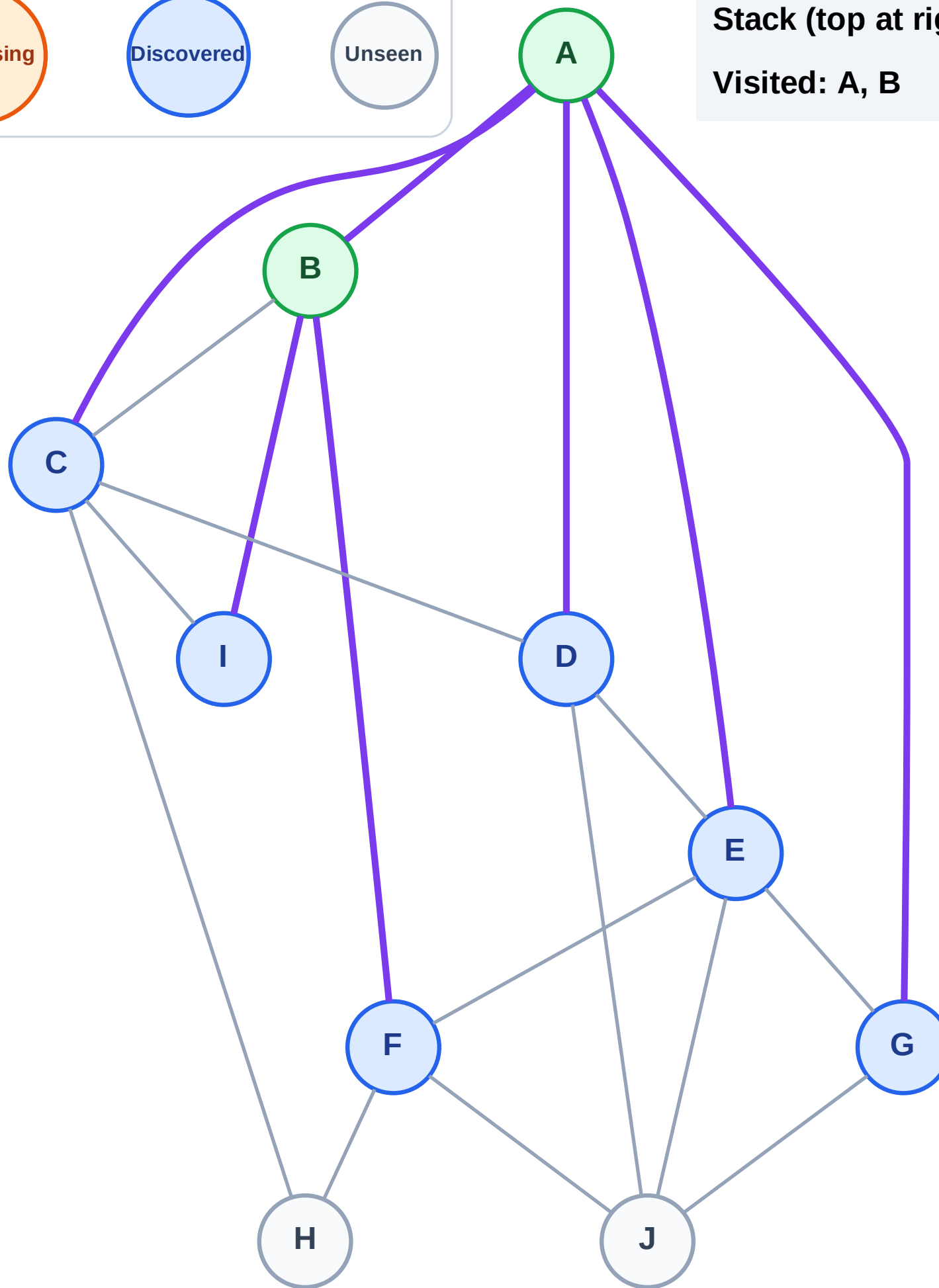


Step 5: Discover I, F from B

Legend

Unseen

Visited: A, B



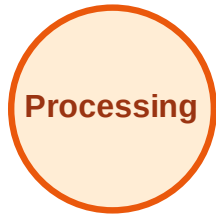
DFS Traversal

Step 6: Process node F

Legend



Complete



Processing



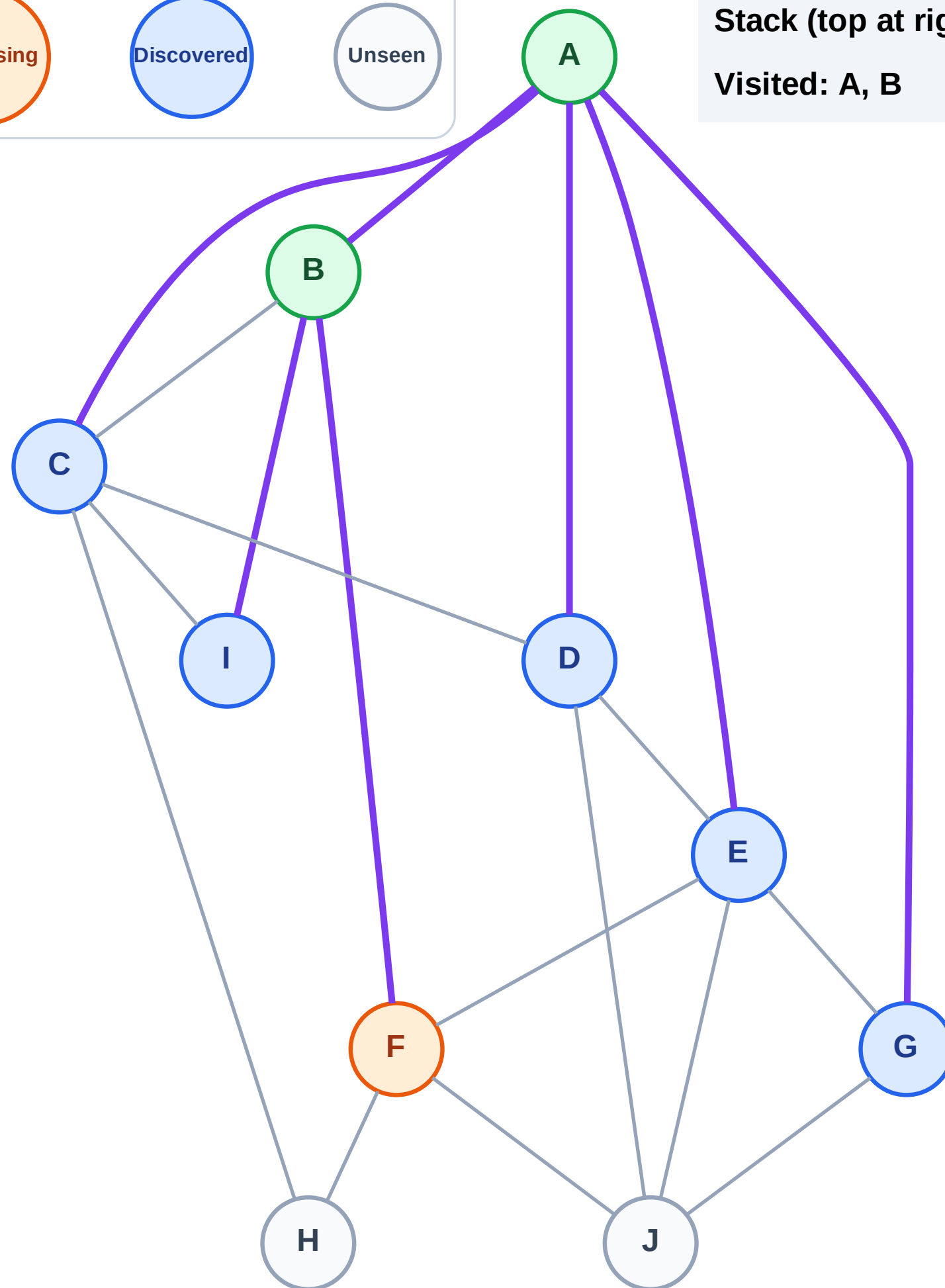
Discovered



Unseen

Stack (top at right): G | E | D | C | I

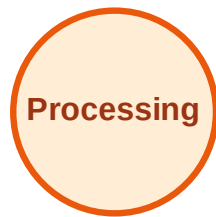
Visited: A, B



DFS Traversal

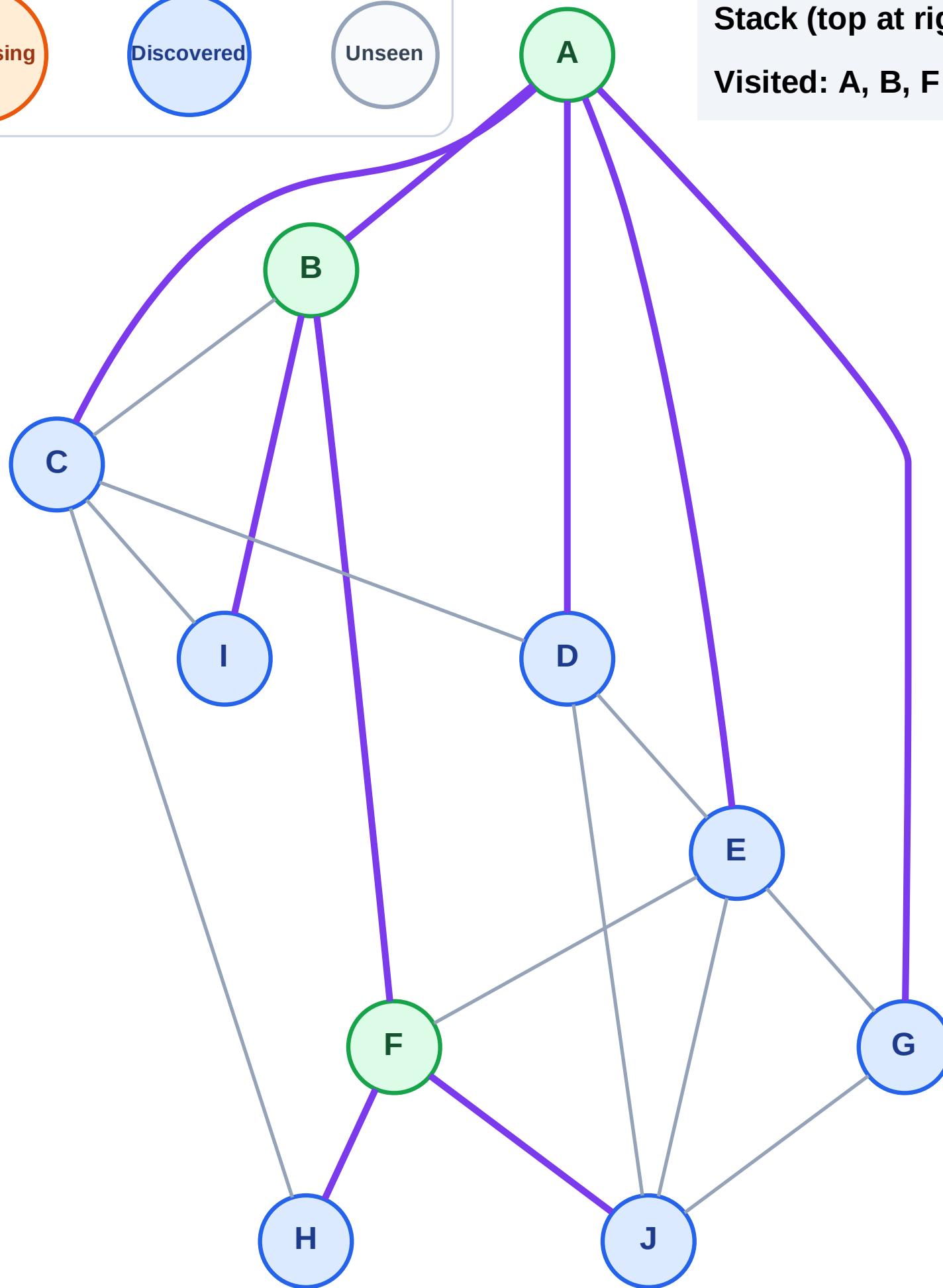
Step 7: Discover J, H from F

Legend



Stack (top at right): G | E | D | C | I | J | H

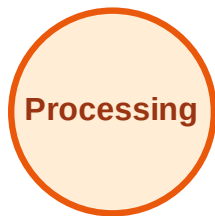
Visited: A, B, F



DFS Traversal

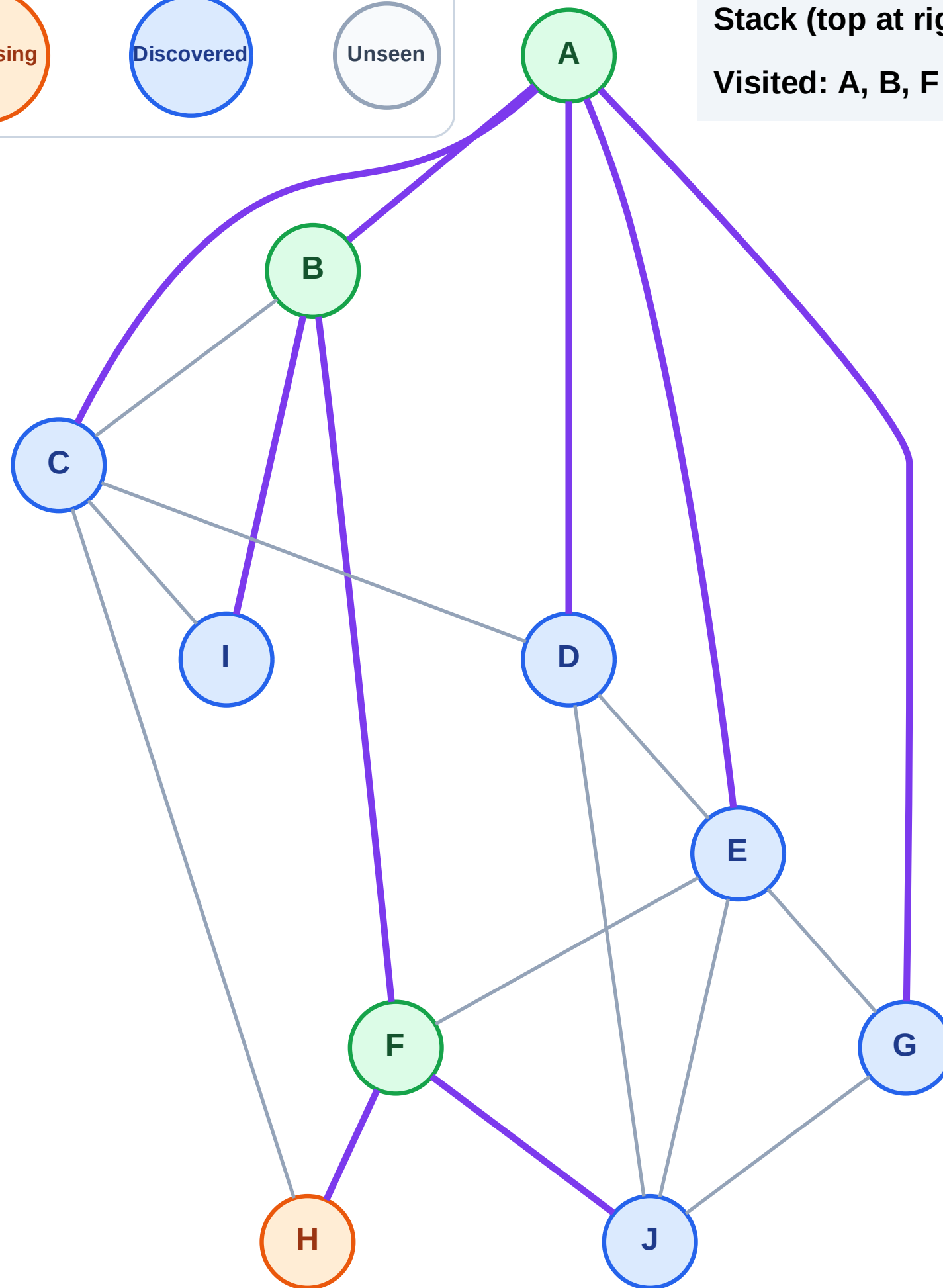
Step 8: Process node H

Legend



Stack (top at right): G | E | D | C | I | J

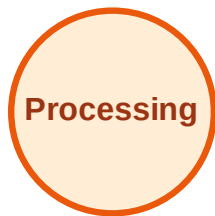
Visited: A, B, F



DFS Traversal

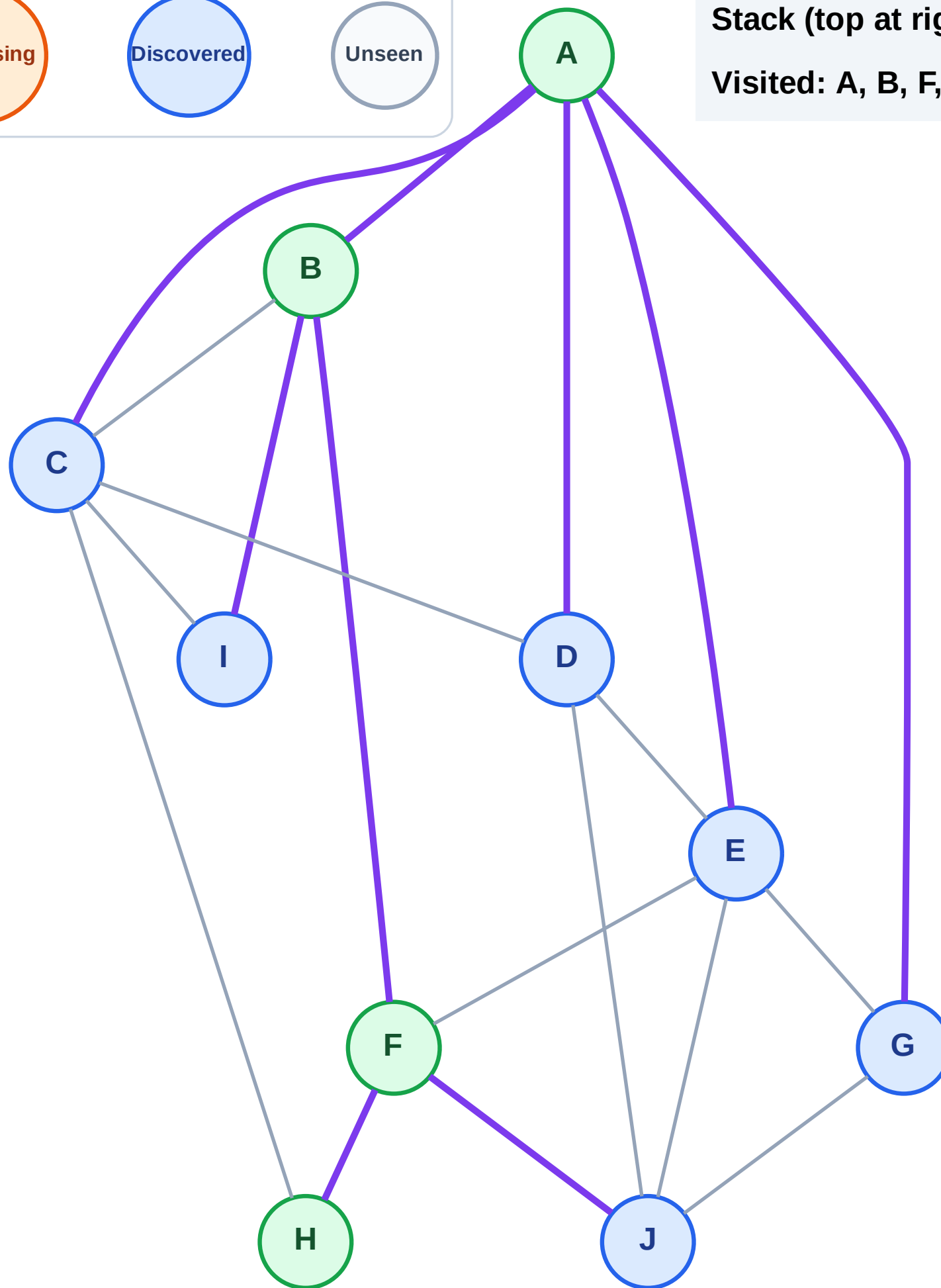
Step 9: No new neighbors from H

Legend



Stack (top at right): G | E | D | C | I | J

Visited: A, B, F, H



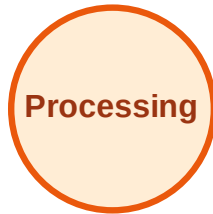
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



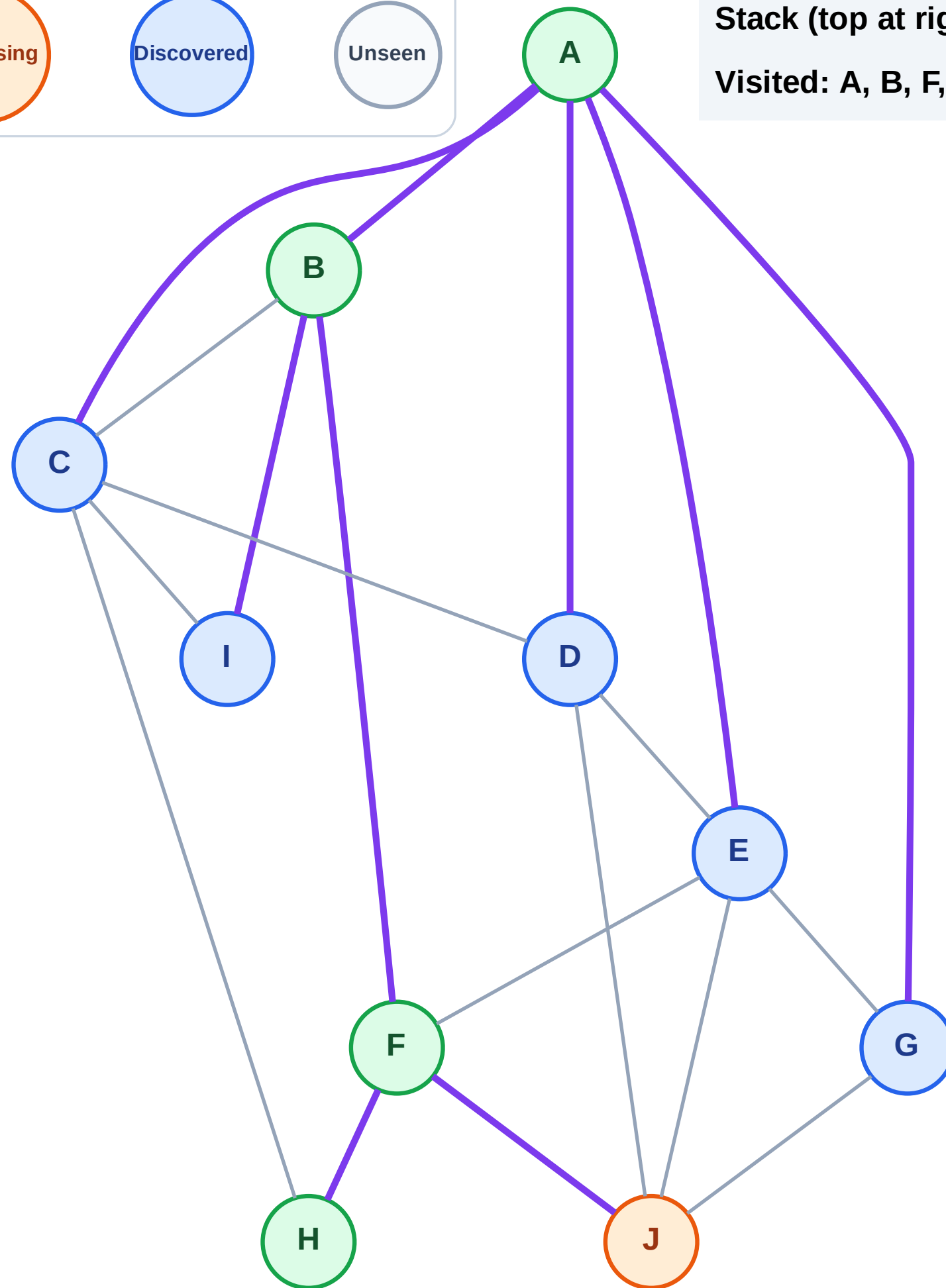
Discovered



Unseen

Stack (top at right): G | E | D | C | I

Visited: A, B, F, H



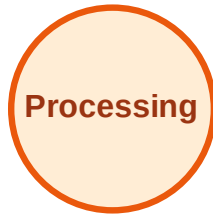
DFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



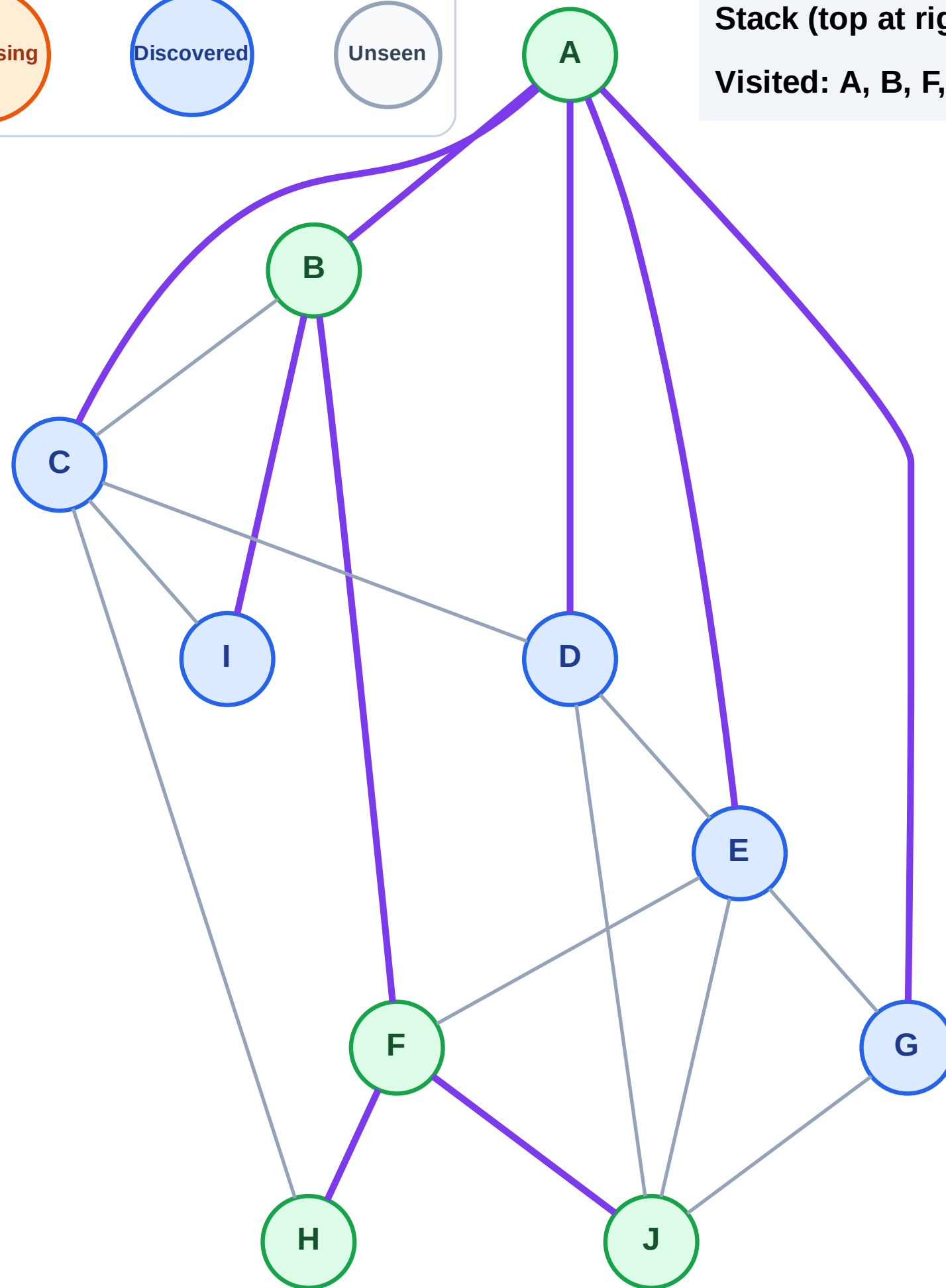
Discovered



Unseen

Stack (top at right): G | E | D | C | I

Visited: A, B, F, H, J



DFS Traversal

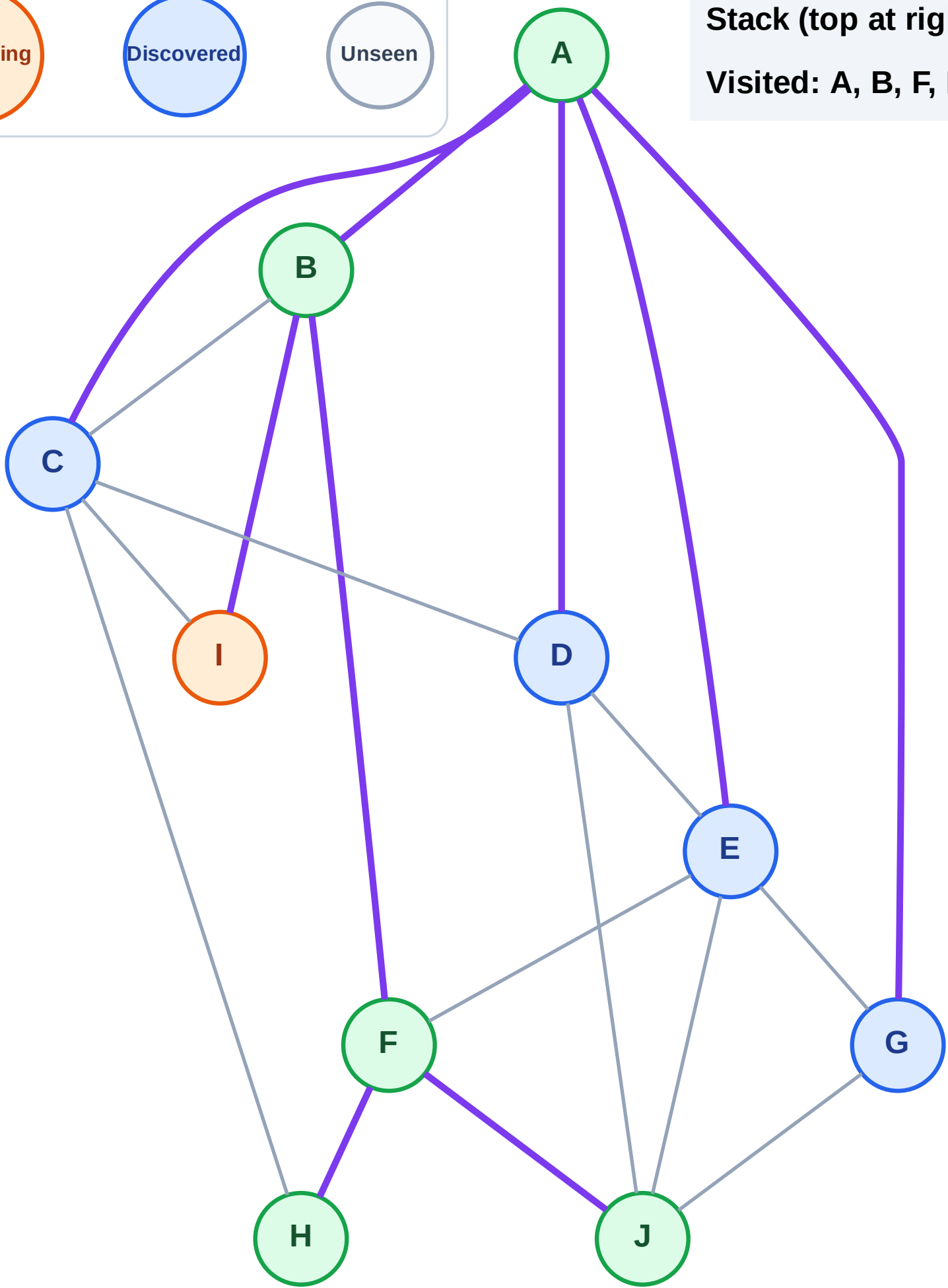
Step 12: Process node I

Legend



Stack (top at right): G | E | D | C

Visited: A, B, F, H, J



DFS Traversal

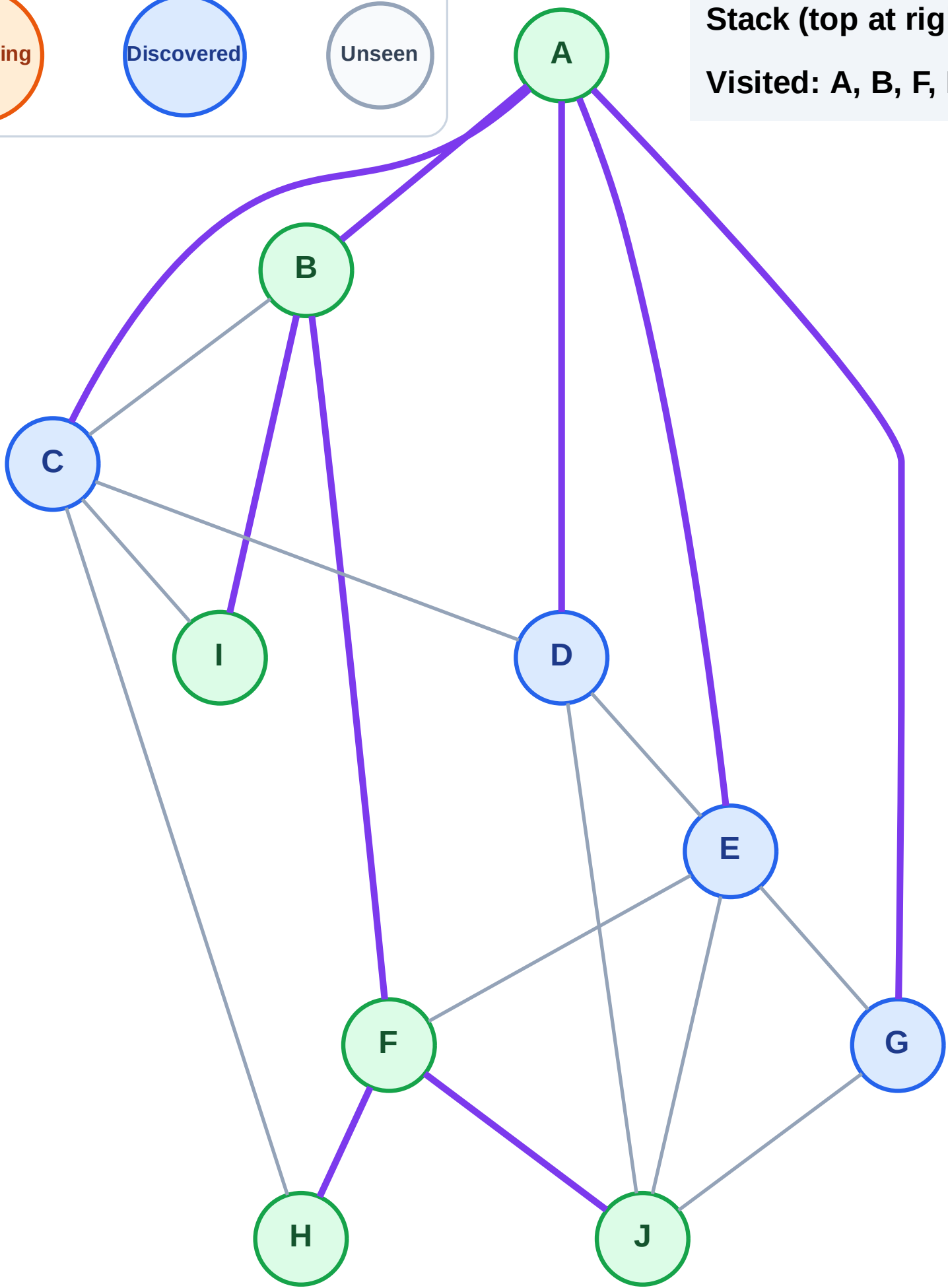
Step 13: No new neighbors from I

Legend



Stack (top at right): G | E | D | C

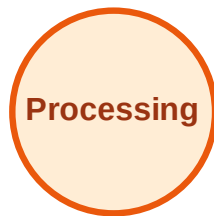
Visited: A, B, F, H, I, J



DFS Traversal

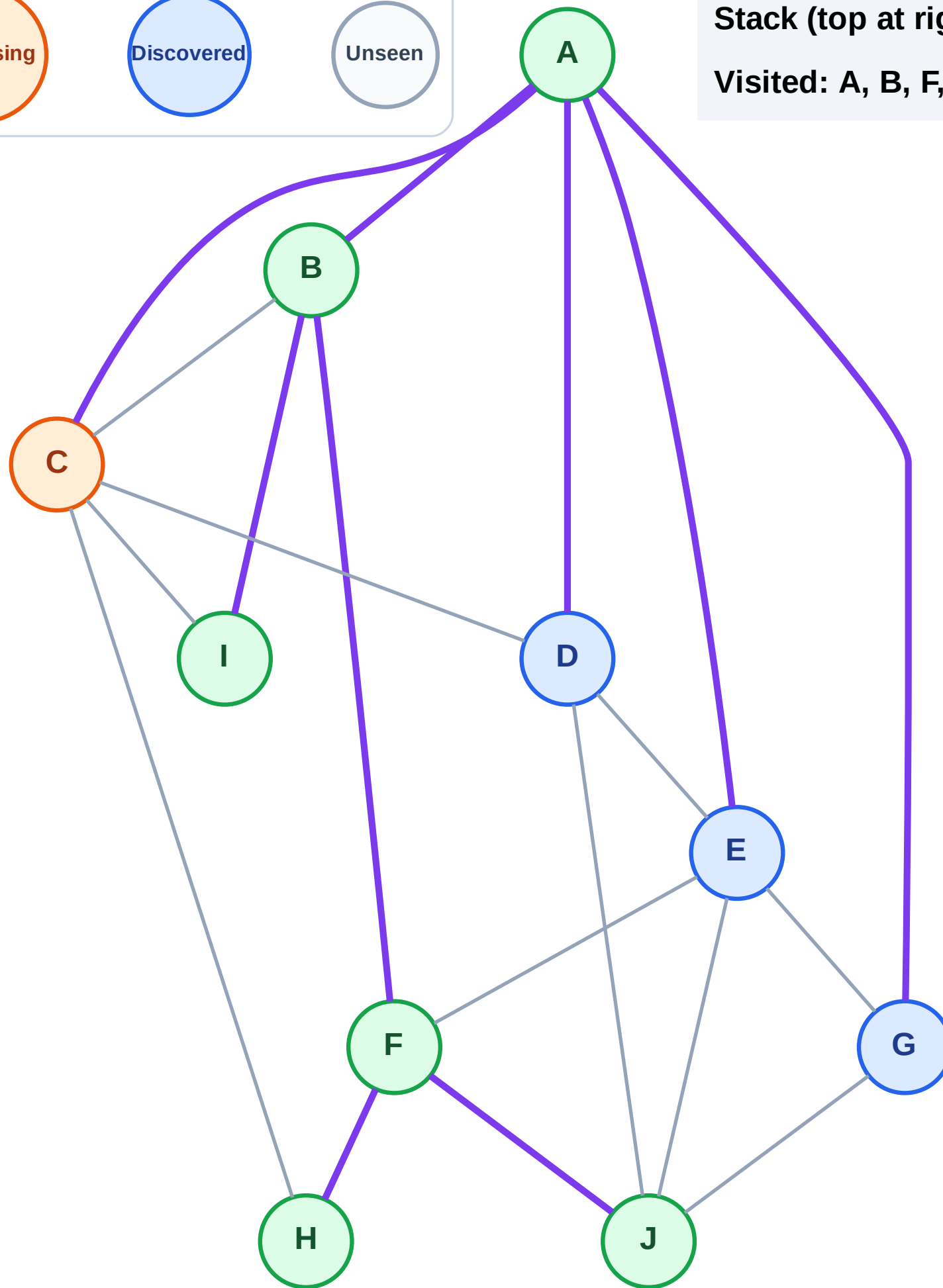
Step 14: Process node C

Legend



Stack (top at right): G | E | D

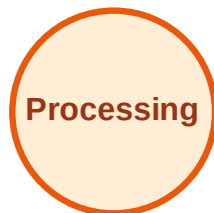
Visited: A, B, F, H, I, J



DFS Traversal

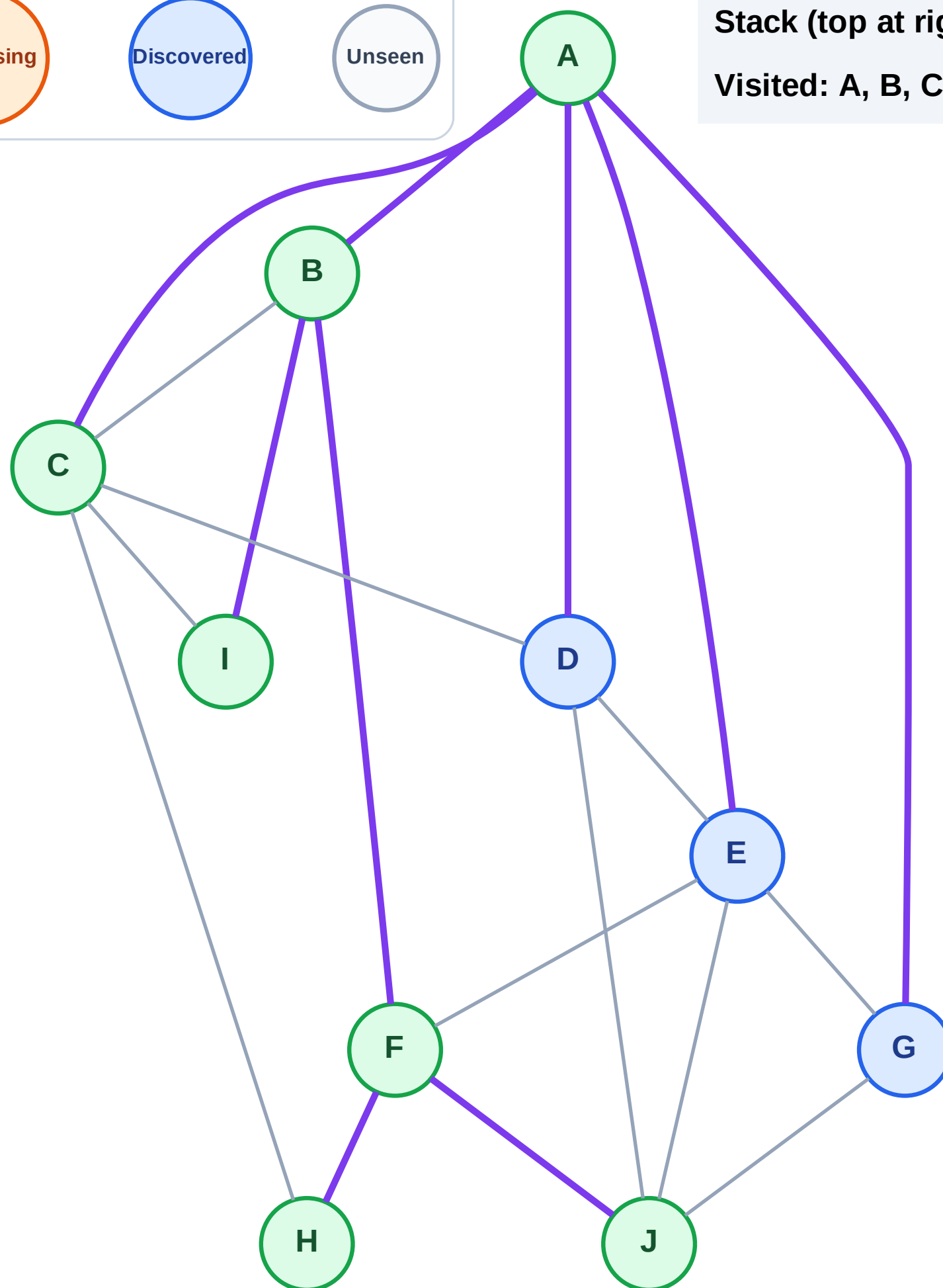
Step 15: No new neighbors from C

Legend



Stack (top at right): G | E | D

Visited: A, B, C, F, H, I, J



DFS Traversal

Step 16: Process node D

Legend

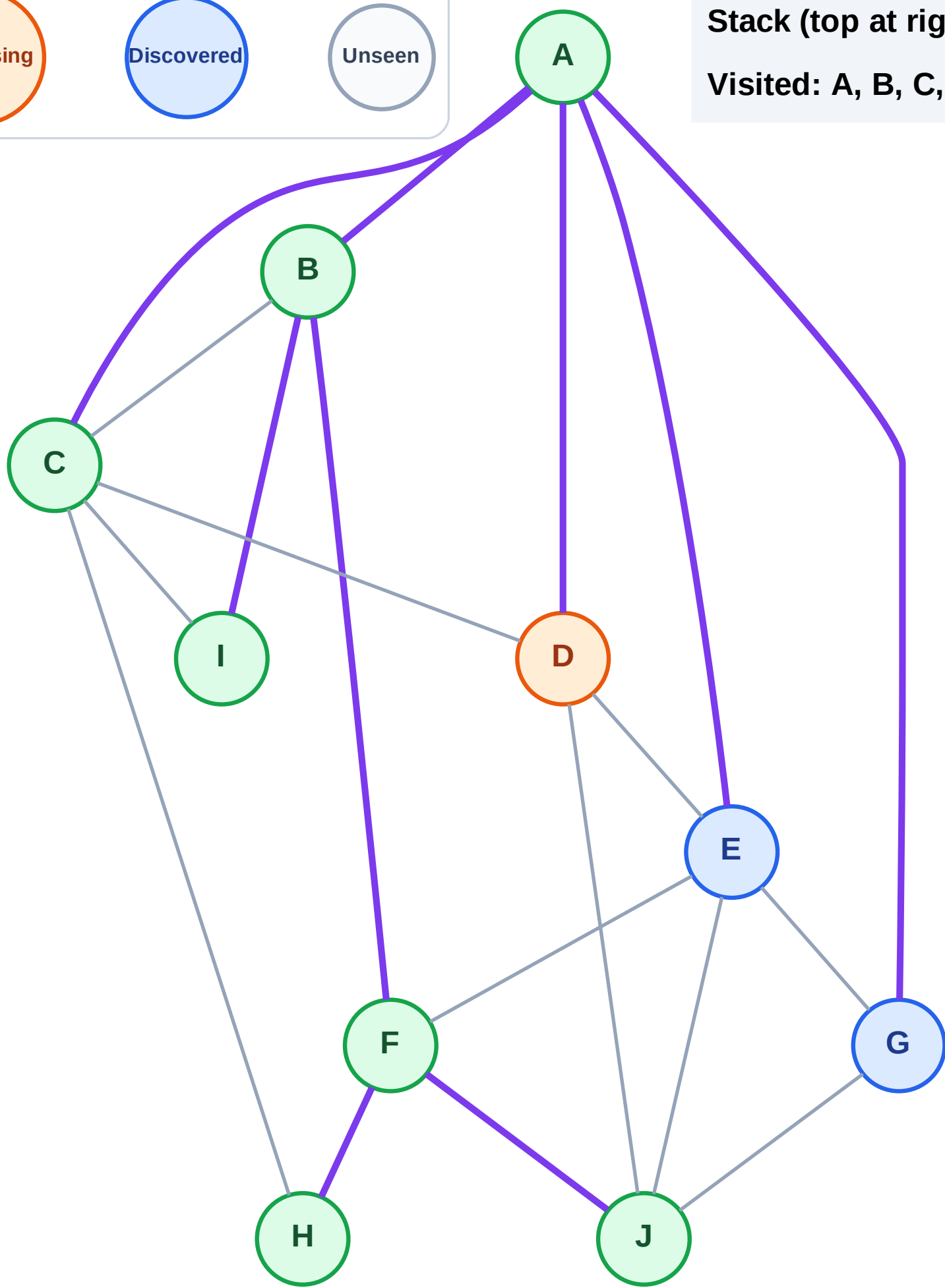
Complete

Processing

Discovered

Unseen

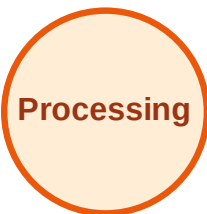
Stack (top at right): G | E
Visited: A, B, C, F, H, I, J



DFS Traversal

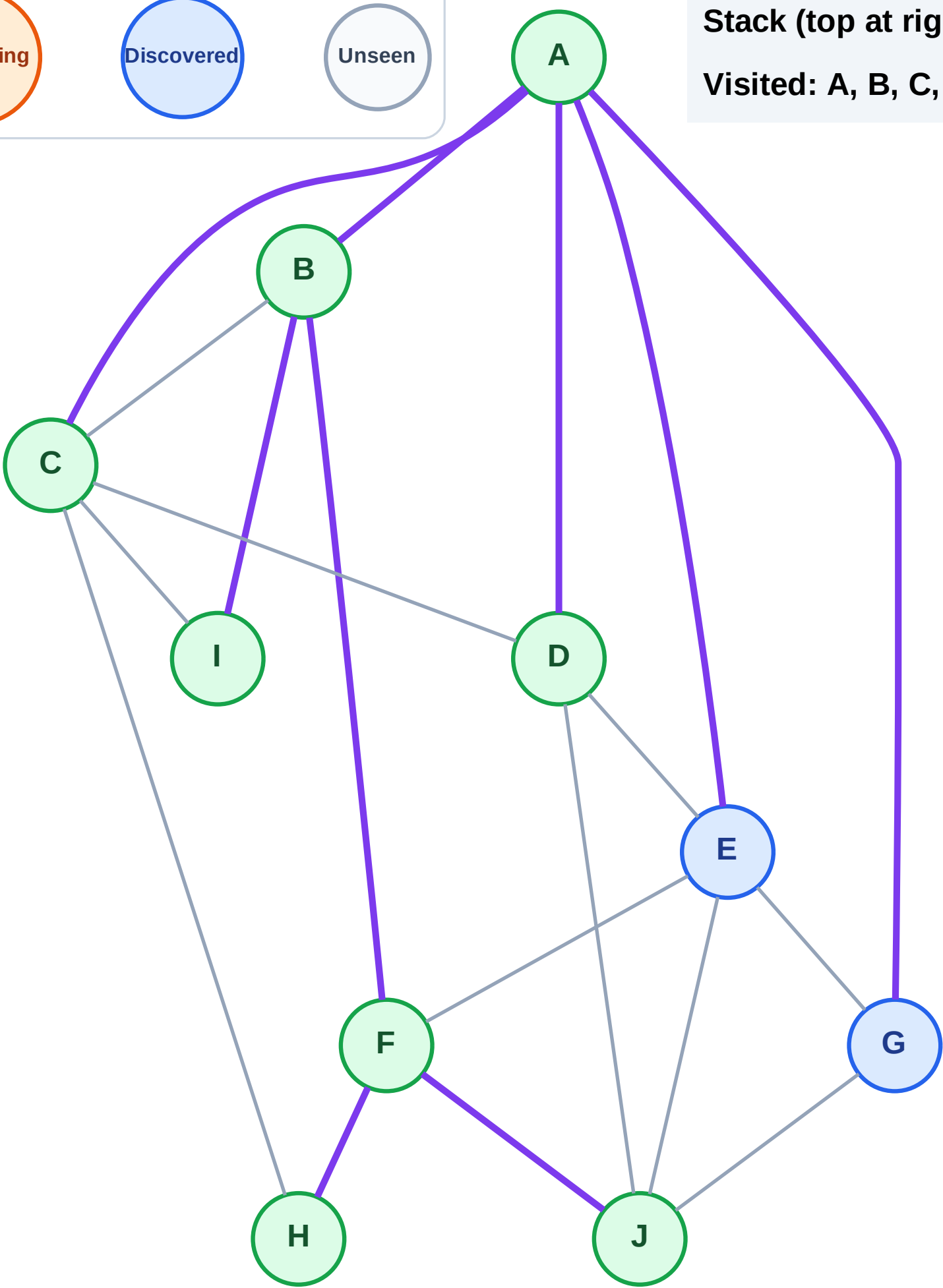
Step 17: No new neighbors from D

Legend



Stack (top at right): G | E

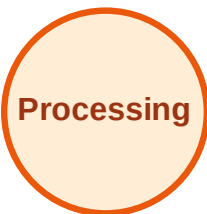
Visited: A, B, C, D, F, H, I, J



DFS Traversal

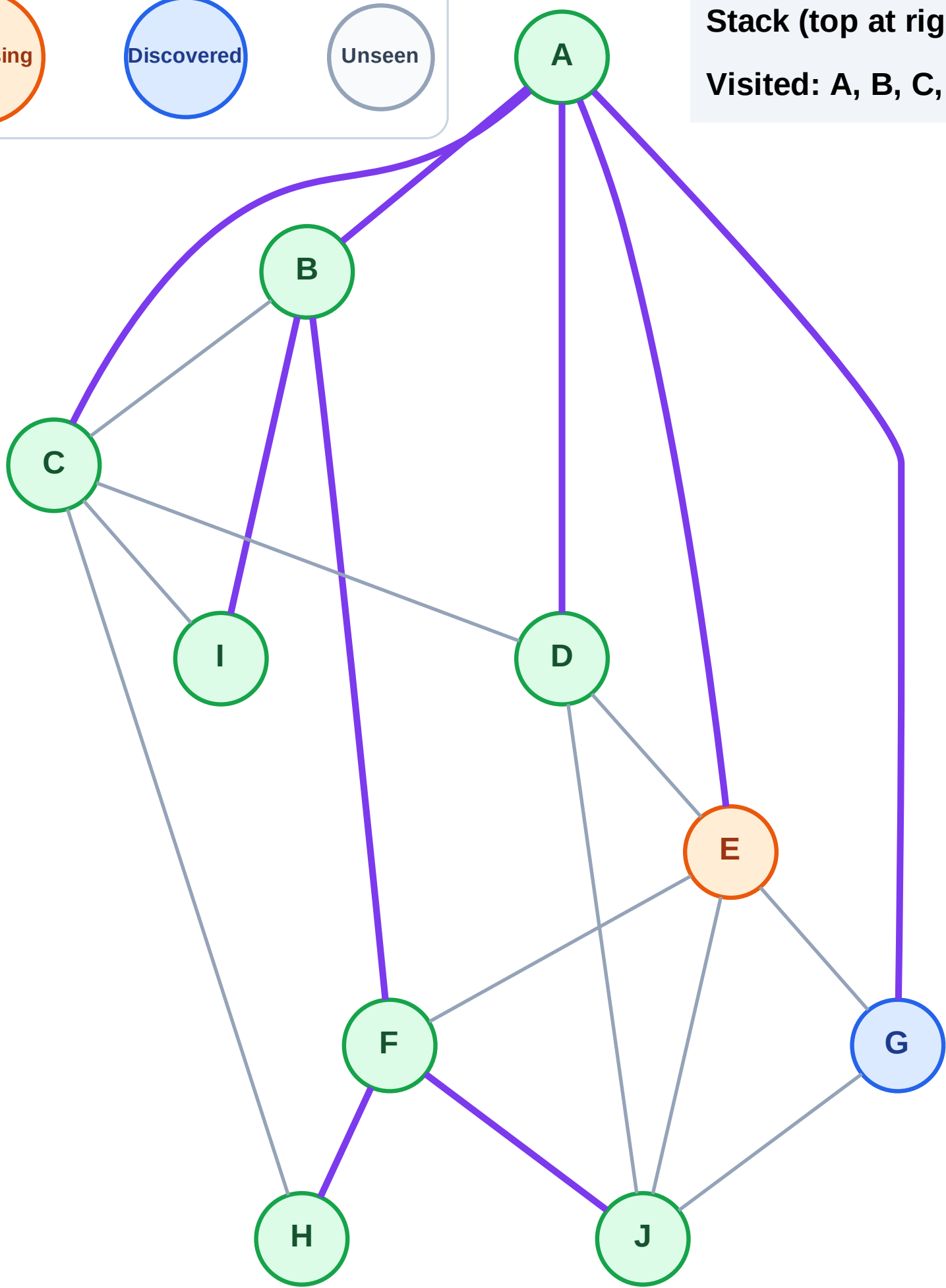
Step 18: Process node E

Legend



Stack (top at right): G

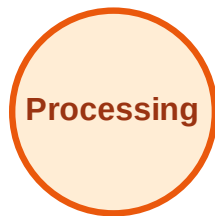
Visited: A, B, C, D, F, H, I, J



DFS Traversal

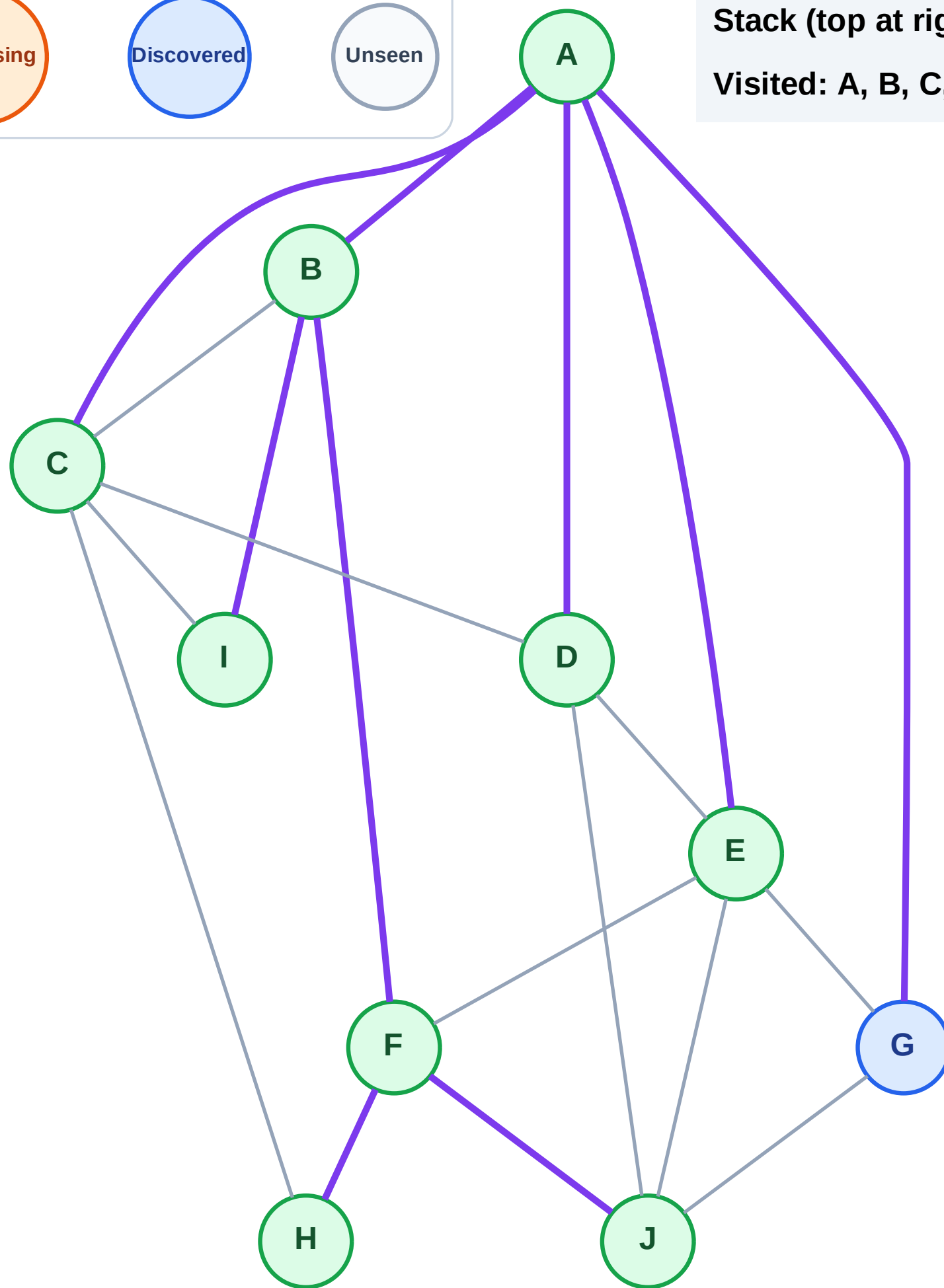
Step 19: No new neighbors from E

Legend

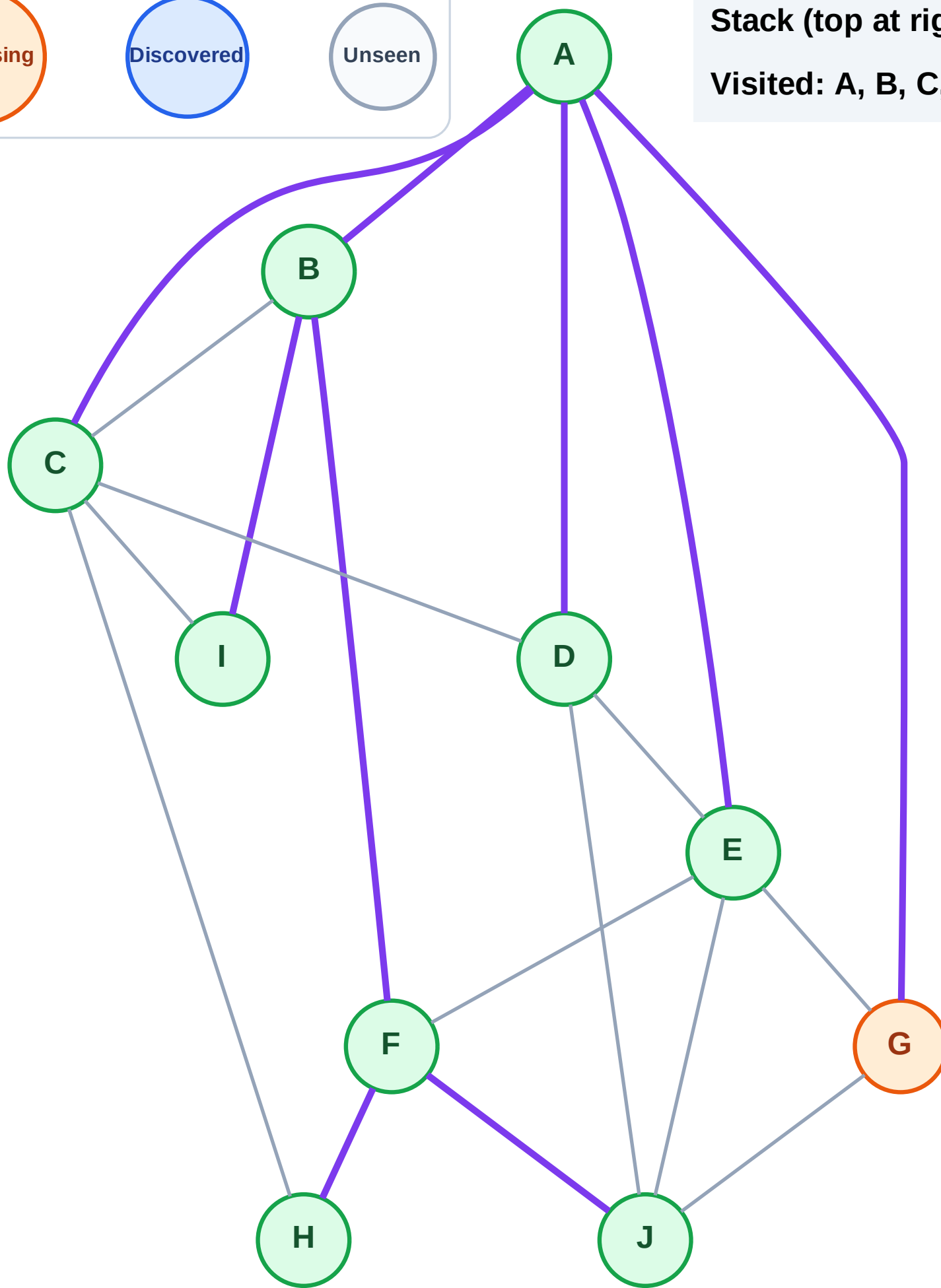
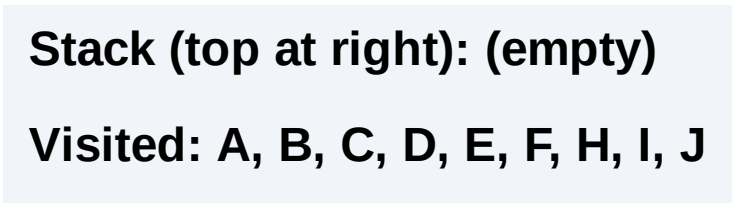


Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



Step 20: Process node G



Step 21: No new neighbors from G

Step 21: No new neighbors from G

